

# Unidad 4

## Estimación de habilidad por pasaje de mensajes



# Unidad 4

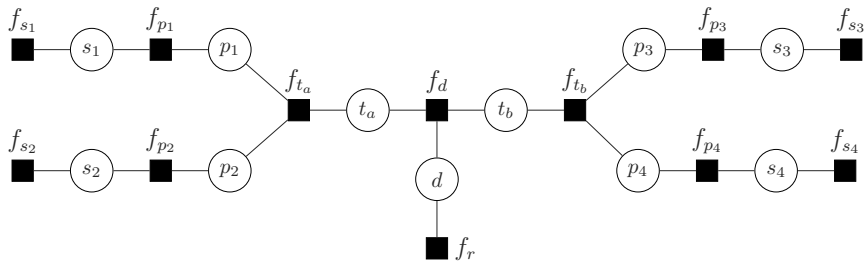
## Estimación de habilidad por pasaje de mensajes

- Herbrich et al. **TrueSkill: A Bayesian Skill Rating System**. Advances in Neural Information Processing Systems. 2006. ([Descargar](#)). Lectura: paper completo.

Más adelante (Unidad 7):

- Dangauthier et al. *Trueskill through time: Revisiting the history of chess*. Advances in Neural Information Processing Systems. 2008. ([Descargar](#)). Lectura: paper completo.

# TrueSkill



$$f_{s_i} = \mathcal{N}(s_i | \mu_i, \sigma^2) \quad f_{p_i} = \mathcal{N}(p_i | s_i, \beta^2) \quad f_{t_e} = \mathbb{I}(t_e = \sum_{i \in A_e} p_i) \quad f_d = \mathbb{I}(d = t_a - t_b) \quad f_r = \mathbb{I}(d > 0)$$

# Propiedades

$$\mathcal{N}(x|\mu, \sigma^2) = \mathcal{N}(\mu|x, \sigma^2) = \mathcal{N}(-\mu|-x, \sigma^2) = \mathcal{N}(-x|-\mu, \sigma^2)$$

# Propiedades

$$\mathcal{N}(x|\mu, \sigma^2) = \mathcal{N}(\mu|x, \sigma^2) = \mathcal{N}(-\mu|-x, \sigma^2) = \mathcal{N}(-x|-\mu, \sigma^2)$$

$$\mathcal{N}(x|\mu, \sigma^2) = \mathcal{N}\left(\frac{X - \mu}{\sigma} | 0, 1\right)$$

# Propiedades

$$\mathcal{N}(x|\mu, \sigma^2) = \mathcal{N}(\mu|x, \sigma^2) = \mathcal{N}(-\mu|-x, \sigma^2) = \mathcal{N}(-x|-\mu, \sigma^2)$$

$$\mathcal{N}(x|\mu, \sigma^2) = \mathcal{N}\left(\frac{X - \mu}{\sigma} | 0, 1\right)$$

$$\frac{\partial}{\partial x} \Phi(x|\mu, \sigma^2) = \mathcal{N}(x|\mu, \sigma^2)$$

# Propiedades

$$\mathcal{N}(x|\mu, \sigma^2) = \mathcal{N}(\mu|x, \sigma^2) = \mathcal{N}(-\mu|-x, \sigma^2) = \mathcal{N}(-x|-\mu, \sigma^2)$$

$$\mathcal{N}(x|\mu, \sigma^2) = \mathcal{N}\left(\frac{X - \mu}{\sigma} | 0, 1\right)$$

$$\frac{\partial}{\partial x} \Phi(x|\mu, \sigma^2) = \mathcal{N}(x|\mu, \sigma^2)$$

$$\int \int_{-\infty}^{\infty} \mathbb{I}(x = h(y, z)) f(x) g(y) dx dy = \int_{-\infty}^{\infty} f(h(y, z)) g(y) dy$$

# Propiedades

$$\mathcal{N}(x|\mu, \sigma^2) = \mathcal{N}(\mu|x, \sigma^2) = \mathcal{N}(-\mu|-x, \sigma^2) = \mathcal{N}(-x|-\mu, \sigma^2)$$

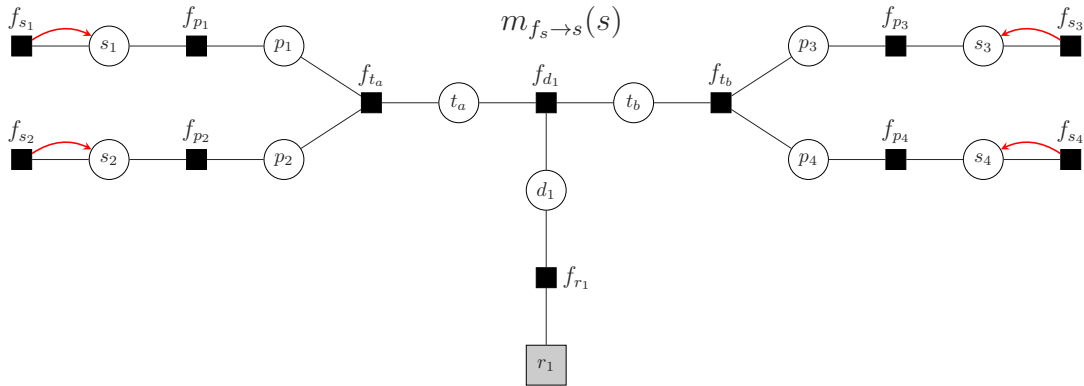
$$\mathcal{N}(x|\mu, \sigma^2) = \mathcal{N}\left(\frac{X - \mu}{\sigma} | 0, 1\right)$$

$$\frac{\partial}{\partial x} \Phi(x|\mu, \sigma^2) = \mathcal{N}(x|\mu, \sigma^2)$$

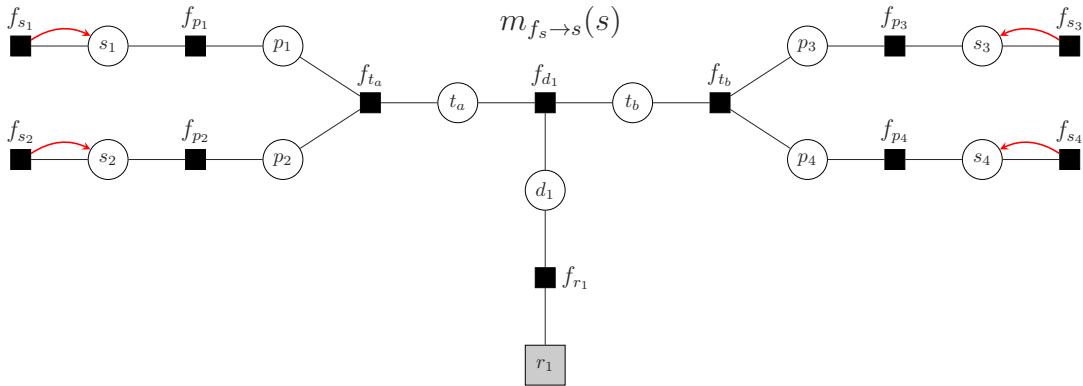
$$\iint_{-\infty}^{\infty} \mathbb{I}(x = h(y, z)) f(x) g(y) dx dy = \int_{-\infty}^{\infty} f(h(y, z)) g(y) dy$$

$$\int_{-\infty}^{\infty} N(x|\mu_x, \sigma_x^2) N(x|\mu_y, \sigma_y^2) dx \stackrel{*}{=} \int_{-\infty}^{\infty} \underbrace{N(\mu_x|\mu_y, \sigma_x^2 + \sigma_y^2)}_{\text{constante}} \underbrace{N(x|\mu_*, \sigma_*^2)}_{\text{integral 1}} dx$$

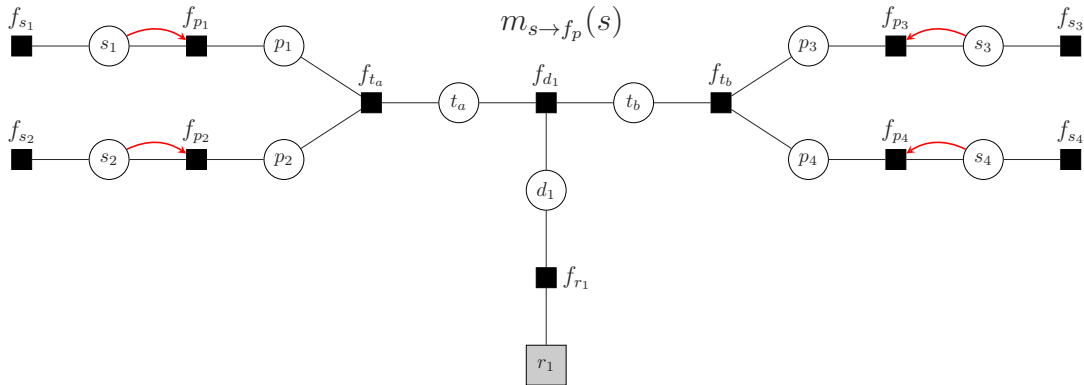




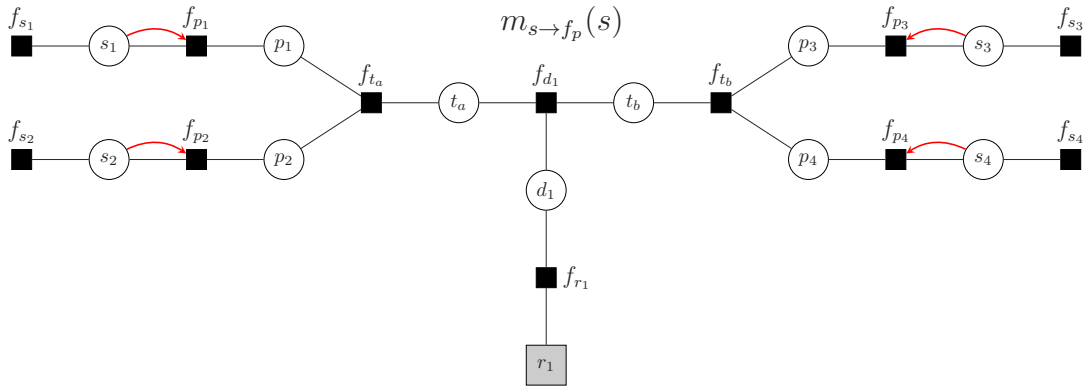
$$m_{f_{s_i} \rightarrow s_i}(s_i) =$$



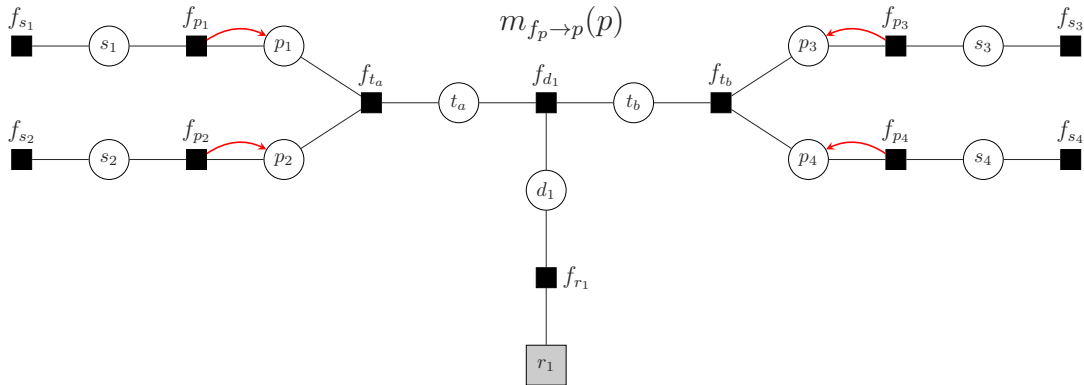
$$m_{f_{s_i} \rightarrow s_i}(s_i) = \mathcal{N}(s_i | \mu_i, \sigma_i^2)$$



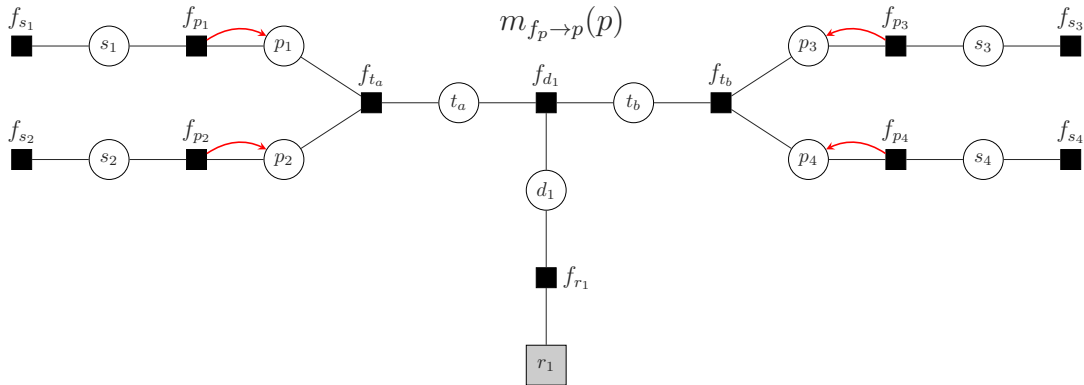
$$m_{s_i \rightarrow f_{p_i}}(s_i) =$$



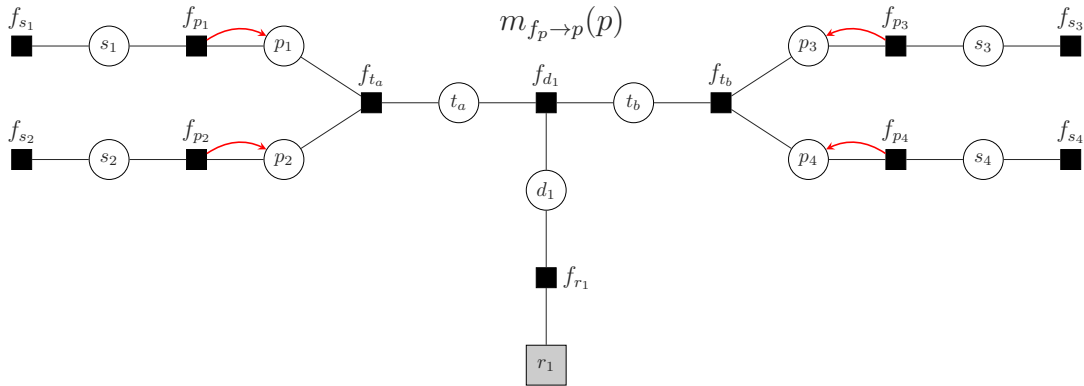
$$m_{s_i \rightarrow f_{p_i}}(s_i) = \mathcal{N}(s_i | \mu_i, \sigma_i^2)$$



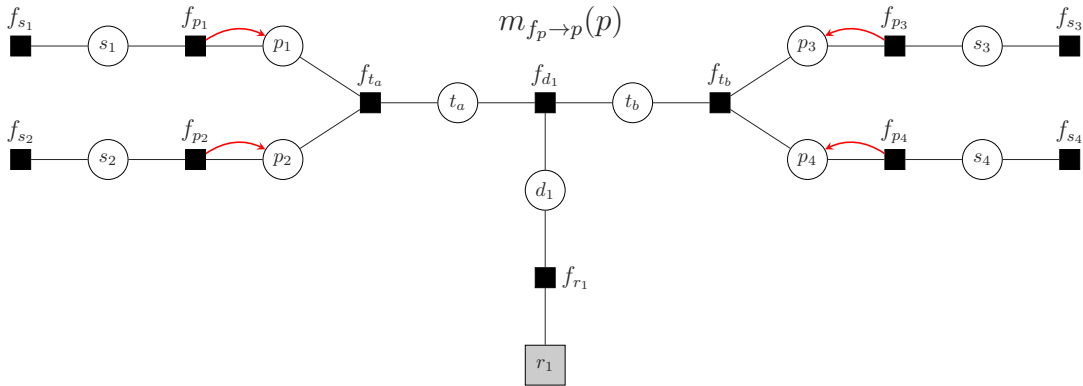
$$m_{f_{p_i} \rightarrow p_i}(p_i) =$$



$$m_{f_{p_i} \rightarrow p_i}(p_i) =$$

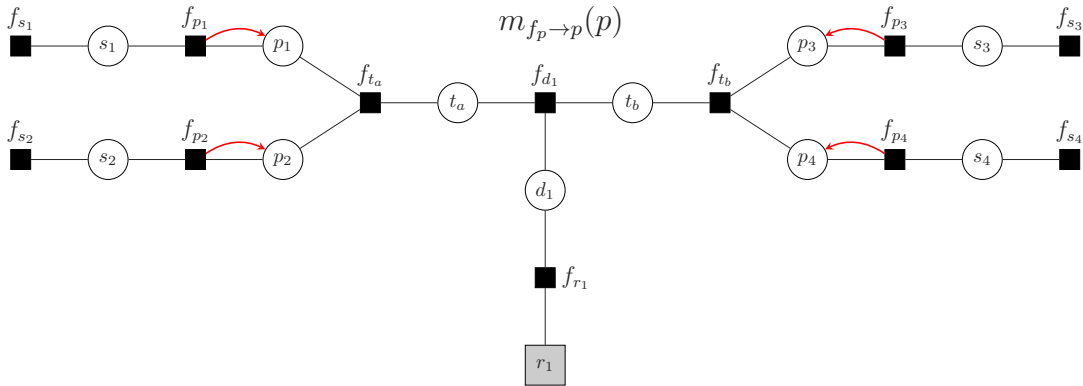


$$m_{f_{p_i} \rightarrow p_i}(p_i) = \int \mathcal{N}(p_i | s_i, \beta^2) \mathcal{N}(s_i | \mu_i, \sigma_i^2) ds_i$$

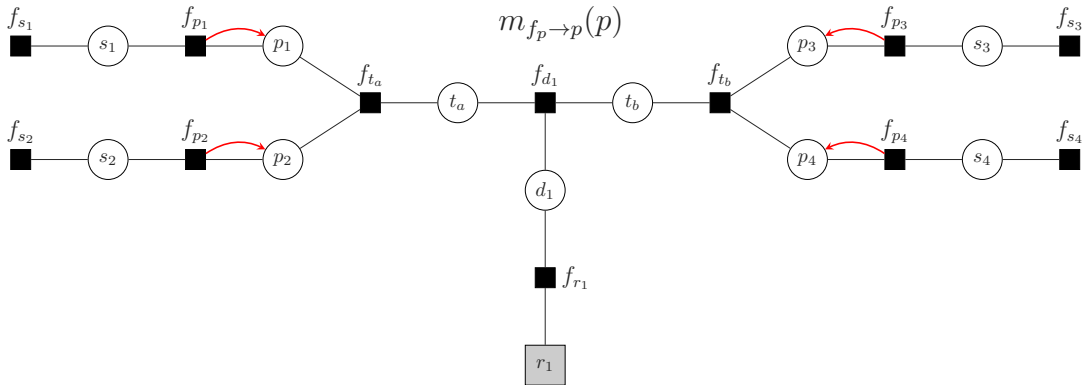


$$m_{f_{p_i} \rightarrow p_i}(p_i) = \int \mathcal{N}(p_i | s_i, \beta^2) \mathcal{N}(s_i | \mu_i, \sigma_i^2) ds_i = \int \mathcal{N}(s_i | p_i, \beta^2) \mathcal{N}(s_i | \mu_i, \sigma_i^2) ds_i$$

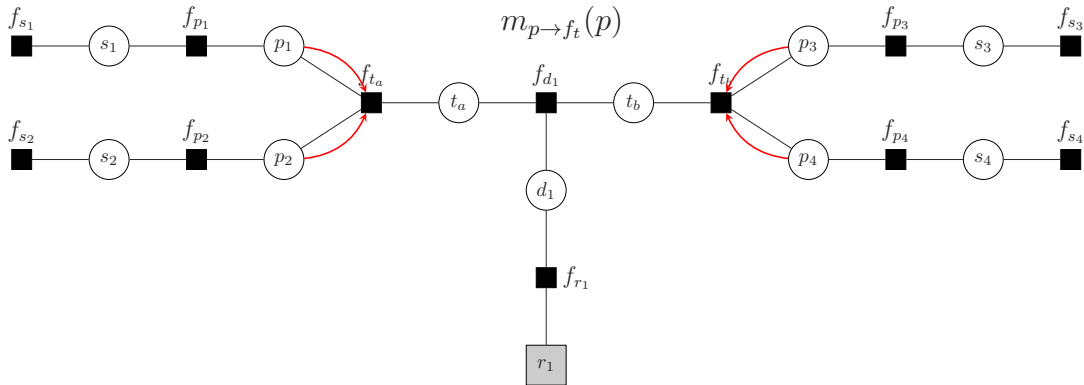




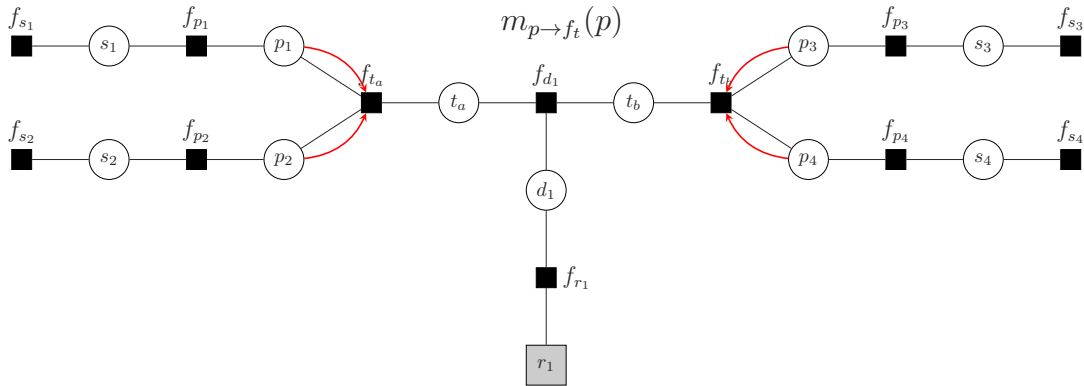
$$\begin{aligned}
 m_{f_{p_i} \rightarrow p_i}(p_i) &= \int \mathcal{N}(p_i | s_i, \beta^2) \mathcal{N}(s_i | \mu_i, \sigma_i^2) ds_i = \int \mathcal{N}(s_i | p_i, \beta^2) \mathcal{N}(s_i | \mu_i, \sigma_i^2) ds_i \\
 &= \int \underbrace{\mathcal{N}(p_i | \mu_i, \beta^2 + \sigma_i^2)}_{\text{const.}} \underbrace{\mathcal{N}(s_i | \mu_*, \sigma_*^2)}_1 ds_i
 \end{aligned}$$



$$\begin{aligned}
 m_{f_{p_i} \rightarrow p_i}(p_i) &= \int \mathcal{N}(p_i | s_i, \beta^2) \mathcal{N}(s_i | \mu_i, \sigma_i^2) ds_i = \int \mathcal{N}(s_i | p_i, \beta^2) \mathcal{N}(s_i | \mu_i, \sigma_i^2) ds_i \\
 &= \int \underbrace{\mathcal{N}(p_i | \mu_i, \beta^2 + \sigma_i^2)}_{\text{const.}} \underbrace{\mathcal{N}(s_i | \mu_*, \sigma_*^2)}_1 ds_i = \mathcal{N}(p_i | \mu_i, \beta^2 + \sigma_i^2)
 \end{aligned}$$

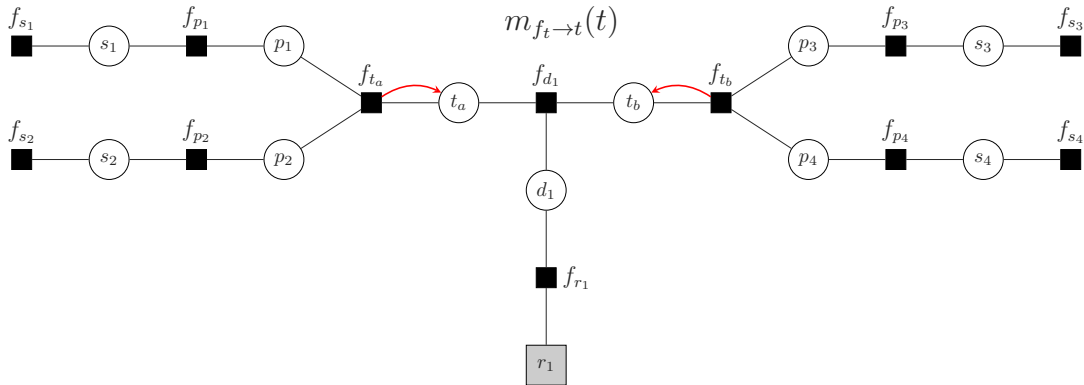


$$m_{p_i \rightarrow f_{t_e}}(p_i) =$$

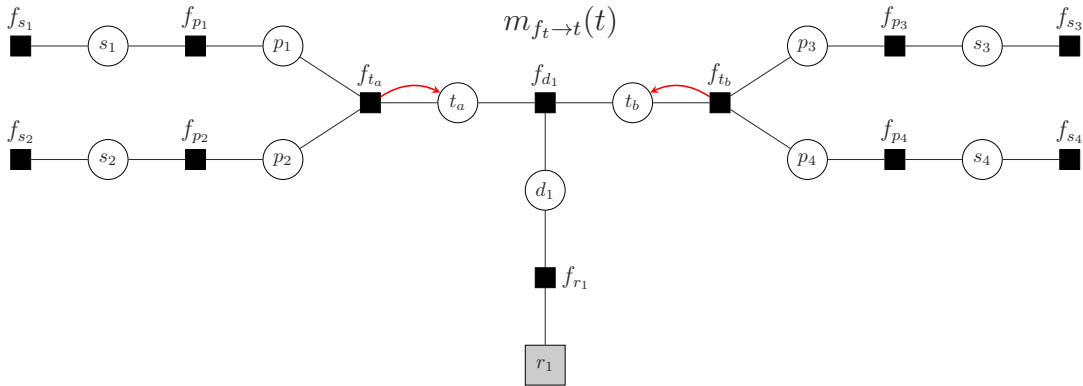


$$m_{p \rightarrow f_t}(p)$$

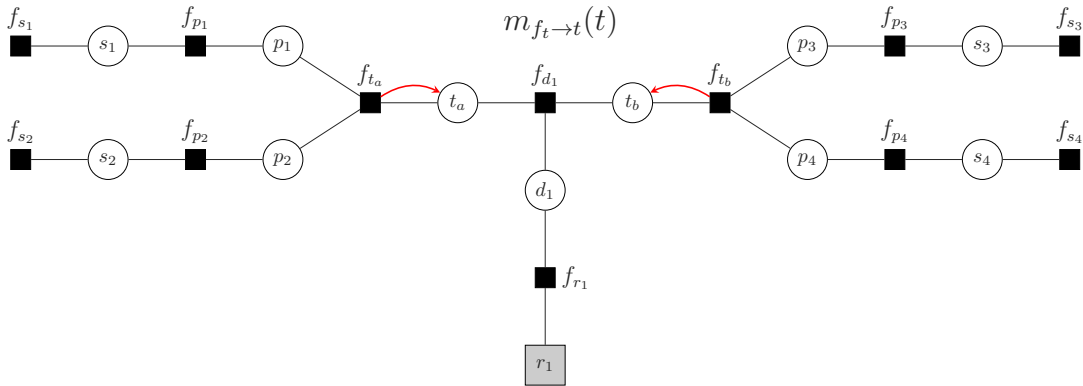
$$m_{p_i \rightarrow f_{t_e}}(p_i) = \mathcal{N}(p_i | \mu_i, \beta^2 + \sigma_i^2)$$



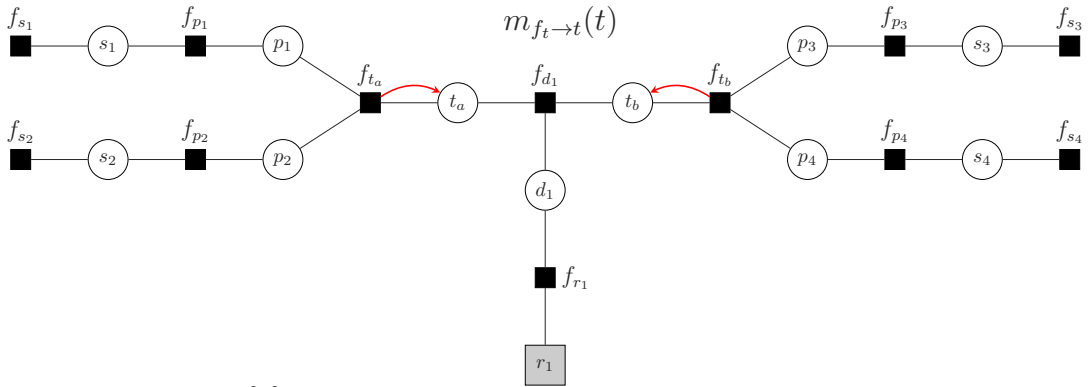
$$m_{f_{t_e} \rightarrow t_e}(t_e) =$$



$$m_{f_{t_e} \rightarrow t_e}(t_e) = \iint \mathbb{I}(t_e = p_i + p_j) \mathcal{N}(p_i | \mu_i, \beta^2 + \sigma_i^2) \mathcal{N}(p_j | \mu_j, \beta^2 + \sigma_j^2) dp_i dp_j$$

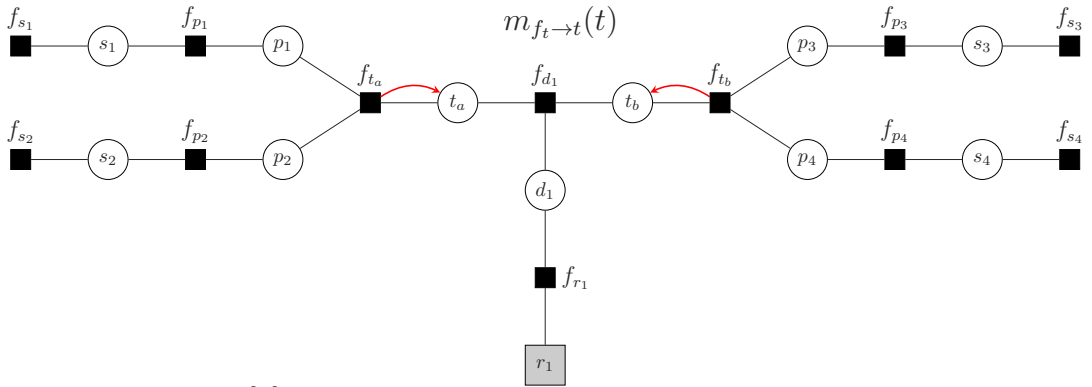


$$\begin{aligned}
 m_{f_{t_e} \rightarrow t_e}(t_e) &= \iint \mathbb{I}(t_e = p_i + p_j) \mathcal{N}(p_i | \mu_i, \beta^2 + \sigma_i^2) \mathcal{N}(p_j | \mu_j, \beta^2 + \sigma_j^2) dp_i dp_j \\
 &= \int \mathcal{N}(p_i | \mu_i, \beta^2 + \sigma_i^2) \mathcal{N}(t_e - p_i | \mu_j, \beta^2 + \sigma_j^2) dp_i
 \end{aligned}$$

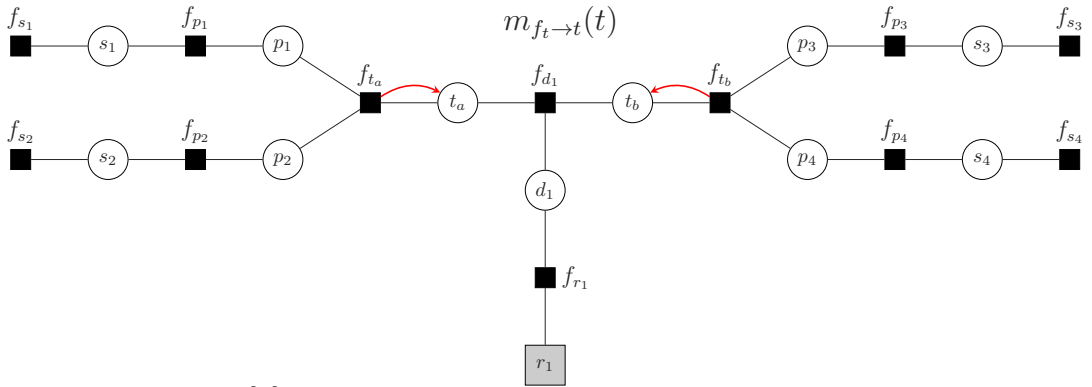


$$\begin{aligned}
 m_{f_{t_e} \rightarrow t_e}(t_e) &= \iint \mathbb{I}(t_e = p_i + p_j) \mathcal{N}(p_i | \mu_i, \beta^2 + \sigma_i^2) \mathcal{N}(p_j | \mu_j, \beta^2 + \sigma_j^2) dp_i dp_j \\
 &= \int \mathcal{N}(p_i | \mu_i, \beta^2 + \sigma_i^2) \mathcal{N}(t_e - p_i | \mu_j, \beta^2 + \sigma_j^2) dp_i \\
 &= \int \underbrace{\mathcal{N}(t_e | \mu_i + \mu_j, 2\beta^2 + \sigma_i^2 + \sigma_j^2)}_{\text{const.}} \underbrace{\mathcal{N}(p_i | \mu_*, \sigma_*^2)}_1 dp_i
 \end{aligned}$$

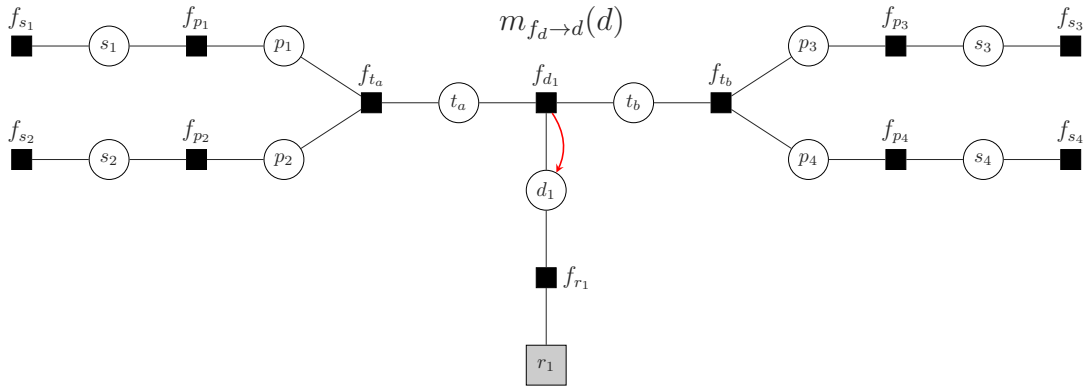




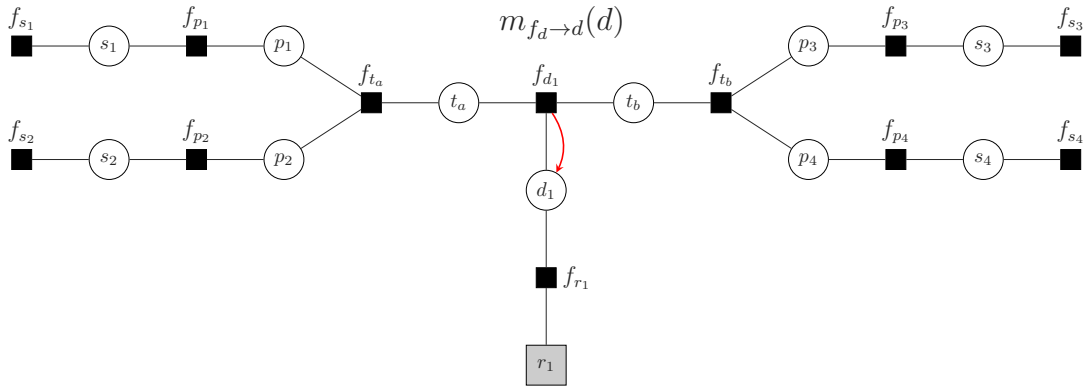
$$\begin{aligned}
 m_{f_{t_e} \rightarrow t_e}(t_e) &= \iint \mathbb{I}(t_e = p_i + p_j) \mathcal{N}(p_i | \mu_i, \beta^2 + \sigma_i^2) \mathcal{N}(p_j | \mu_j, \beta^2 + \sigma_j^2) dp_i dp_j \\
 &= \int \mathcal{N}(p_i | \mu_i, \beta^2 + \sigma_i^2) \mathcal{N}(t_e - p_i | \mu_j, \beta^2 + \sigma_j^2) dp_i \\
 &= \mathcal{N}(t_e | \mu_i + \mu_j, 2\beta^2 + \sigma_i^2 + \sigma_j^2)
 \end{aligned}$$



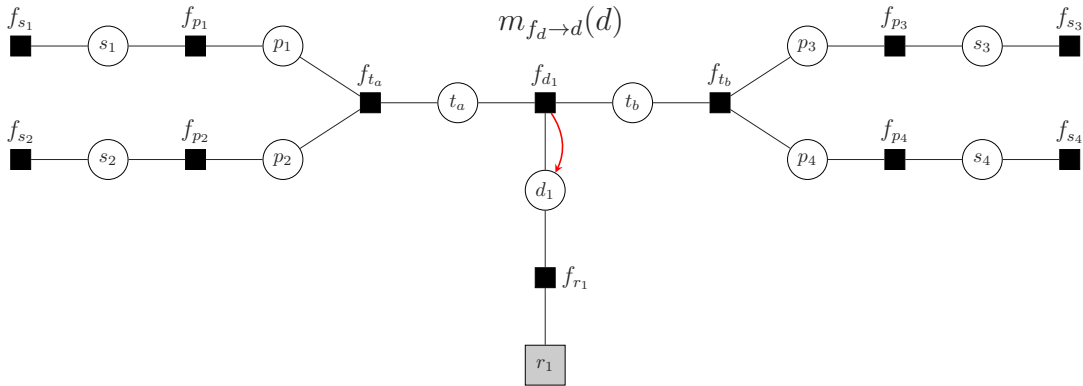
$$\begin{aligned}
 m_{f_{t_e} \rightarrow t_e}(t_e) &= \iint \mathbb{I}(t_e = p_i + p_j) \mathcal{N}(p_i | \mu_i, \beta^2 + \sigma_i^2) \mathcal{N}(p_j | \mu_j, \beta^2 + \sigma_j^2) dp_i dp_j \\
 &= \int \mathcal{N}(p_i | \mu_i, \beta^2 + \sigma_i^2) \mathcal{N}(t_e - p_i | \mu_j, \beta^2 + \sigma_j^2) dp_i \\
 &= \mathcal{N}(t_e | \underbrace{\mu_i + \mu_j}_{\mu_e}, \underbrace{2\beta^2 + \sigma_i^2 + \sigma_j^2}_{\sigma_e^2})
 \end{aligned}$$



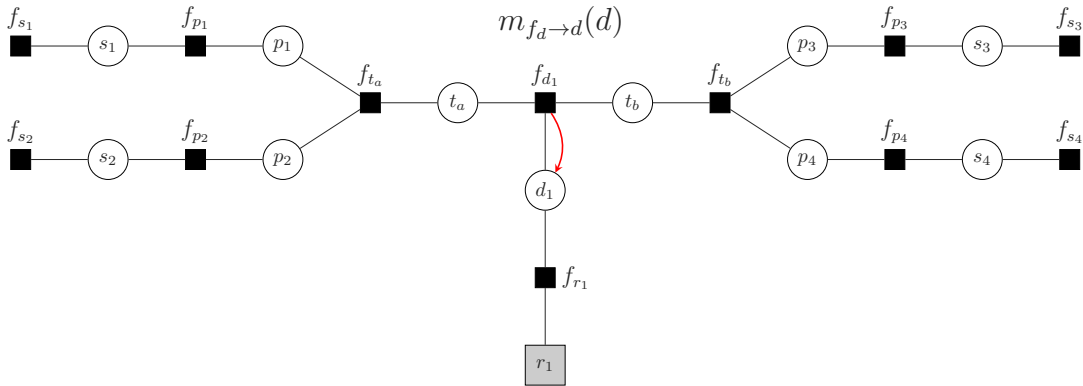
$$m_{f_d \rightarrow d}(d) =$$



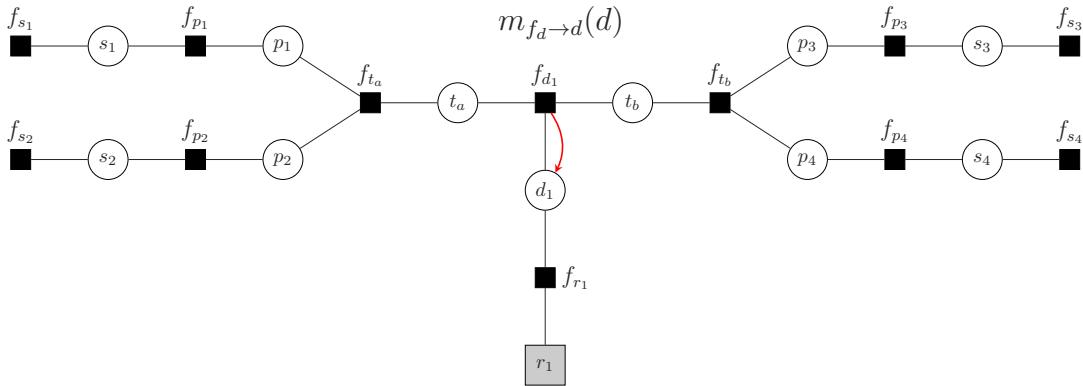
$$m_{f_d \rightarrow d}(d) = \iint \mathbb{I}(d = t_a - t_b) \mathcal{N}(t_a | \mu_a, \sigma_a^2) \mathcal{N}(t_b | \mu_b, \sigma_b^2) dt_a dt_b$$



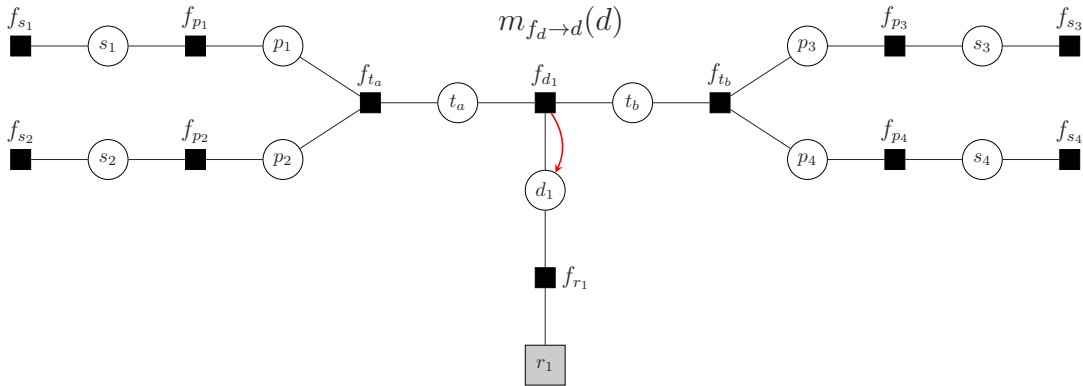
$$\begin{aligned}
 m_{f_d \rightarrow d}(d) &= \iint \mathbb{I}(d = t_a - t_b) \mathcal{N}(t_a | \mu_a, \sigma_a^2) \mathcal{N}(t_b | \mu_b, \sigma_b^2) dt_a dt_b \\
 &= \int \mathcal{N}(d + t_b | \mu_a, \sigma_a^2) \mathcal{N}(t_b | \mu_b, \sigma_b^2) dt_b
 \end{aligned}$$



$$\begin{aligned}
 m_{f_d \rightarrow d}(d) &= \iint \mathbb{I}(d = t_a - t_b) \mathcal{N}(t_a | \mu_a, \sigma_a^2) \mathcal{N}(t_b | \mu_b, \sigma_b^2) dt_a dt_b \\
 &= \int \mathcal{N}(t_b | \mu_a - d, \sigma_a^2) \mathcal{N}(t_b | \mu_b, \sigma_b^2) dt_b
 \end{aligned}$$

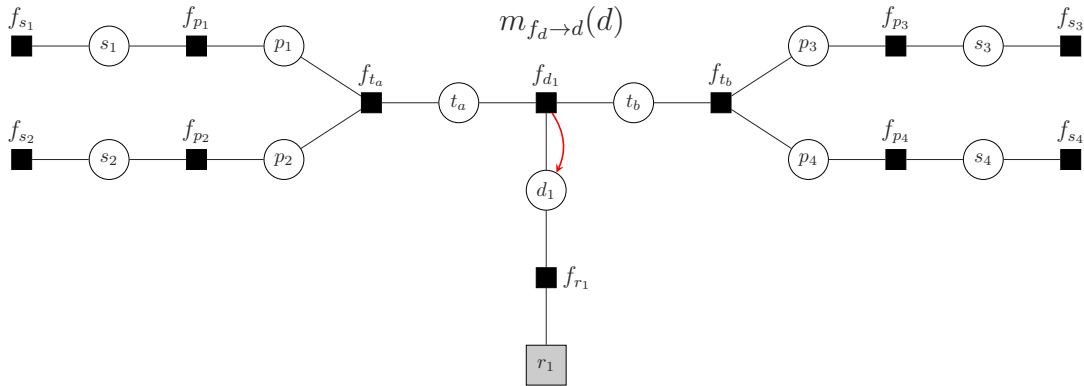


$$\begin{aligned}
 m_{f_d \rightarrow d}(d) &= \iint \mathbb{I}(d = t_a - t_b) \mathcal{N}(t_a | \mu_a, \sigma_a^2) \mathcal{N}(t_b | \mu_b, \sigma_b^2) dt_a dt_b \\
 &= \int \mathcal{N}(t_b | \mu_a - d, \sigma_a^2) \mathcal{N}(t_b | \mu_b, \sigma_b^2) dt_b \\
 &= \int \underbrace{\mathcal{N}(d | \mu_a - \mu_b, \sigma_a^2 + \sigma_b^2)}_{\text{const.}} \underbrace{\mathcal{N}(t_b | \mu_*, \sigma_*^2)}_1 dt_b
 \end{aligned}$$

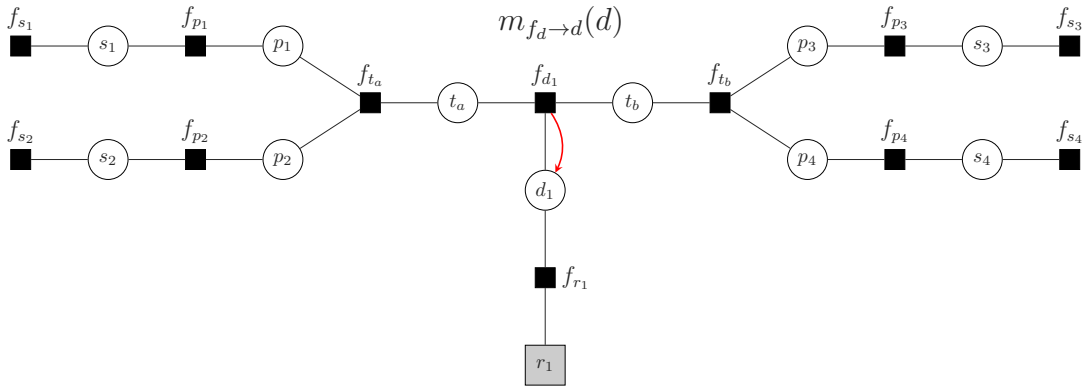


$$\begin{aligned}
 m_{f_d \rightarrow d}(d) &= \iint \mathbb{I}(d = t_a - t_b) \mathcal{N}(t_a | \mu_a, \sigma_a^2) \mathcal{N}(t_b | \mu_b, \sigma_b^2) dt_a dt_b \\
 &= \int \mathcal{N}(t_b | \mu_a - d, \sigma_a^2) \mathcal{N}(t_b | \mu_b, \sigma_b^2) dt_b \\
 &= \mathcal{N}(d | \mu_a - \mu_b, \sigma_a^2 + \sigma_b^2)
 \end{aligned}$$

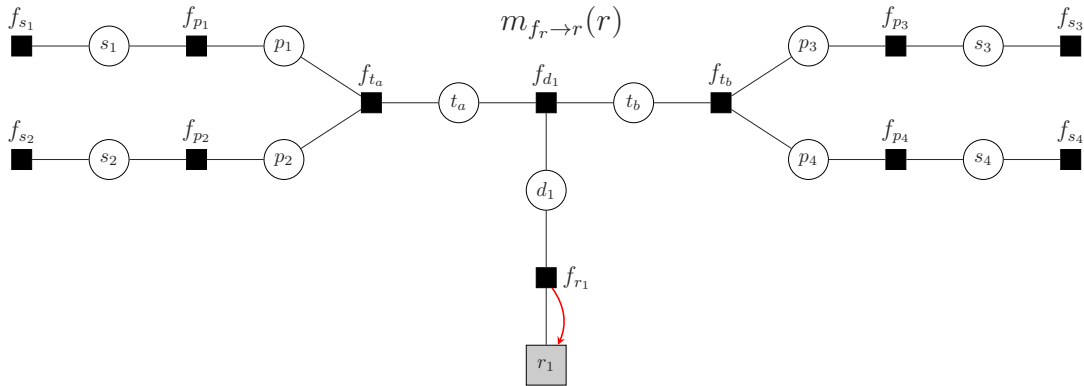




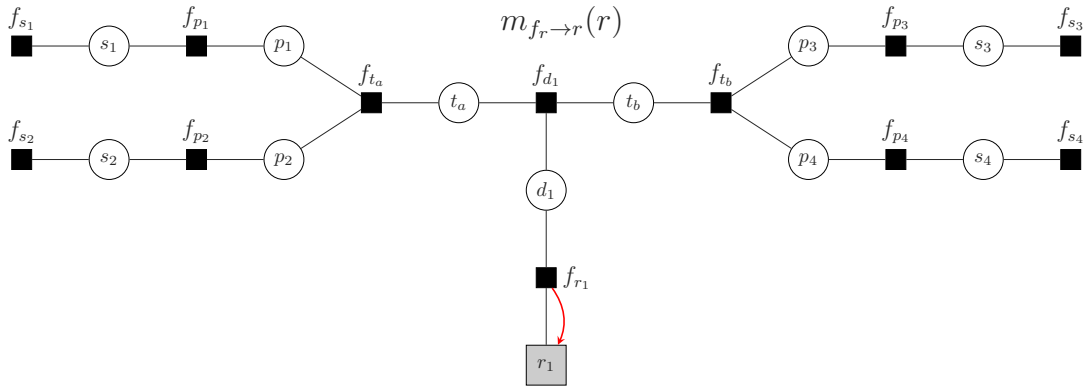
$$\begin{aligned}
 m_{f_d \rightarrow d}(d) &= \iint \mathbb{I}(d = t_a - t_b) \mathcal{N}(t_a | \mu_a, \sigma_a^2) \mathcal{N}(t_b | \mu_b, \sigma_b^2) dt_a dt_b \\
 &= \int \mathcal{N}(t_b | \mu_a - d, \sigma_a^2) \mathcal{N}(t_b | \mu_b, \sigma_b^2) dt_b \\
 &= \mathcal{N}(d | \underbrace{\mu_a - \mu_b}_{\text{Diferencia esperada } \delta}, \underbrace{\sigma_a^2 + \sigma_b^2}_{\text{Varianza total } \vartheta})
 \end{aligned}$$



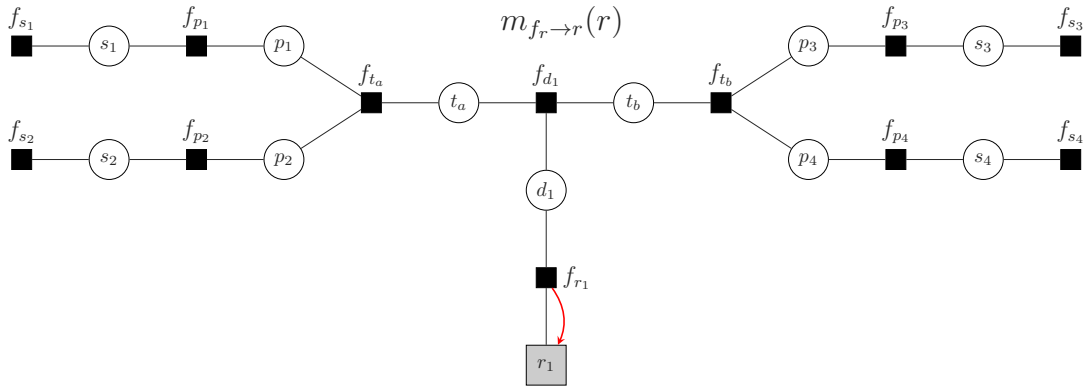
$$\begin{aligned}
 m_{f_d \rightarrow d}(d) &= \iint \mathbb{I}(d = t_a - t_b) \mathcal{N}(t_a | \mu_a, \sigma_a^2) \mathcal{N}(t_b | \mu_b, \sigma_b^2) dt_a dt_b \\
 &= \int \mathcal{N}(t_b | \mu_a - d, \sigma_a^2) \mathcal{N}(t_b | \mu_b, \sigma_b^2) dt_b \\
 &= \mathcal{N}(d | \underbrace{\mu_a - \mu_b}_{\substack{\text{Diferencia } \delta \\ \text{esperada}}}, \underbrace{\sigma_a^2 + \sigma_b^2}_{\substack{\text{Varianza } \vartheta \\ \text{total}}}) = \mathcal{N}(d | \delta, \vartheta^2)
 \end{aligned}$$



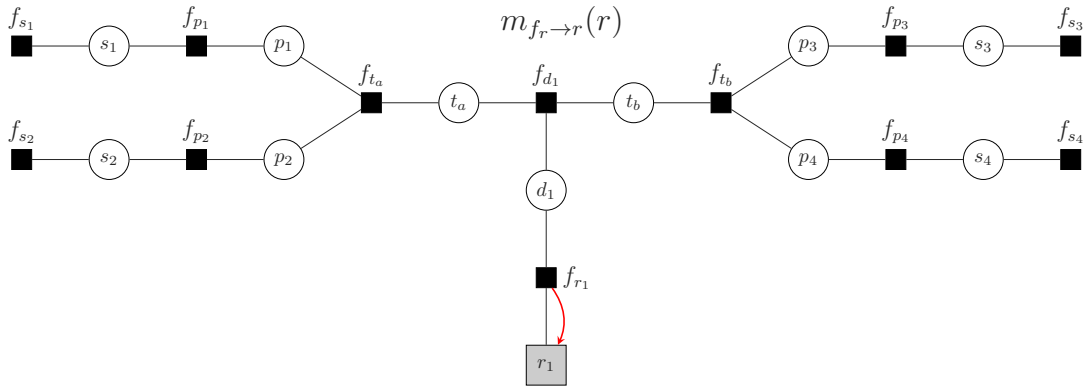
$$m_{f_r \rightarrow r}(r) =$$



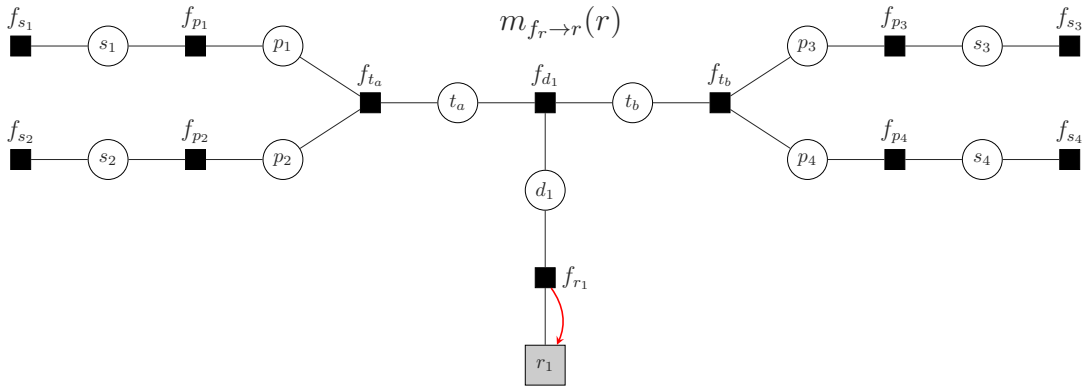
$$m_{f_r \rightarrow r}(r) = \int \mathbb{I}(d > 0) \mathcal{N}(d | \delta, \vartheta^2) dd$$



$$\begin{aligned}
 m_{f_r \rightarrow r}(r) &= \int \mathbb{I}(d > 0) \mathcal{N}(d|\delta, \vartheta^2) dd \\
 &= \int_0^\infty \mathcal{N}(d|\delta, \vartheta^2) dd
 \end{aligned}$$



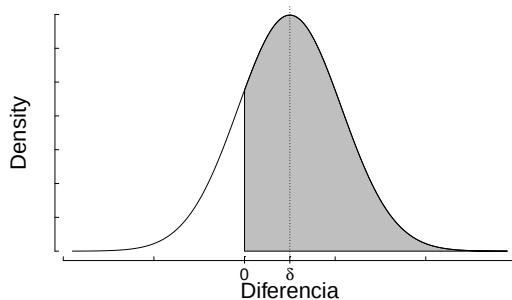
$$\begin{aligned}
 m_{f_r \rightarrow r}(r) &= \int \mathbb{I}(d > 0) \mathcal{N}(d|\delta, \vartheta^2) dd \\
 &= \int_0^\infty \mathcal{N}(d|\delta, \vartheta^2) dd \\
 &= 1 - \Phi(0|\delta, \vartheta^2)
 \end{aligned}$$



$$\begin{aligned}
 m_{f_r \rightarrow r}(r) &= \int \mathbb{I}(d > 0) \mathcal{N}(d|\delta, \vartheta^2) dd \\
 &= \int_0^\infty \mathcal{N}(d|\delta, \vartheta^2) dd \\
 &= 1 - \Phi(0|\delta, \vartheta^2) = \Phi\left(\frac{\delta}{\vartheta}\right)
 \end{aligned}$$

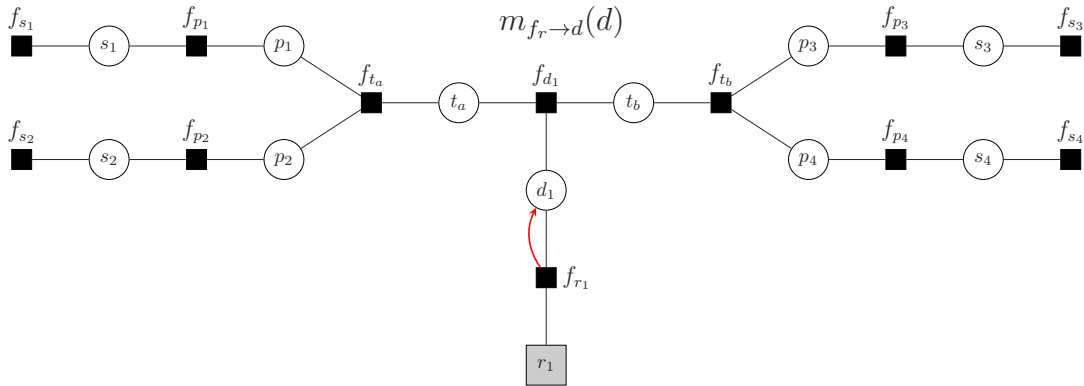
## Evidencia

$$P(\text{Resultado}|\text{Modelo}) = 1 - \Phi\left(0 \mid \underbrace{(\overbrace{\mu_1 + \mu_2}^{\mu_a}) - (\overbrace{\mu_3 + \mu_4}^{\mu_b})}_{\text{Diferencia esperada: } \delta}, \underbrace{\overbrace{2\beta^2 + \sigma_1^2 + \sigma_2^2}^{\sigma_a^2} + \overbrace{2\beta^2 + \sigma_3^2 + \sigma_4^2}^{\sigma_b^2}}_{\text{incertidumbre total: } \vartheta^2}\right)$$

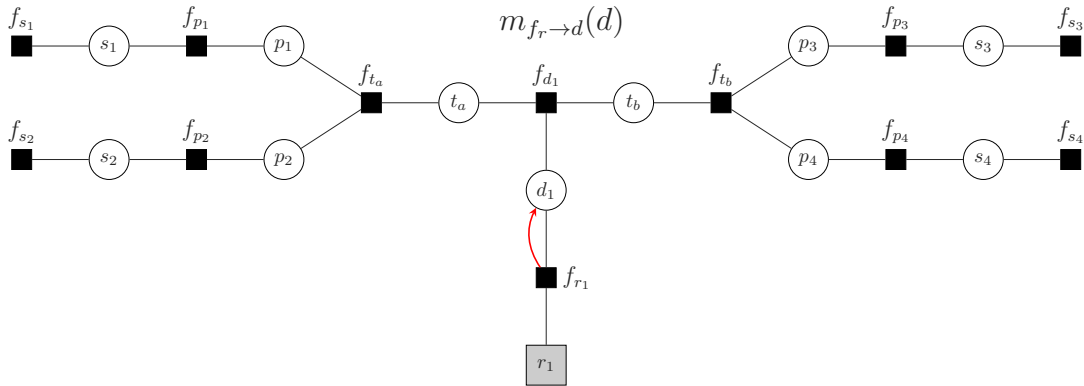




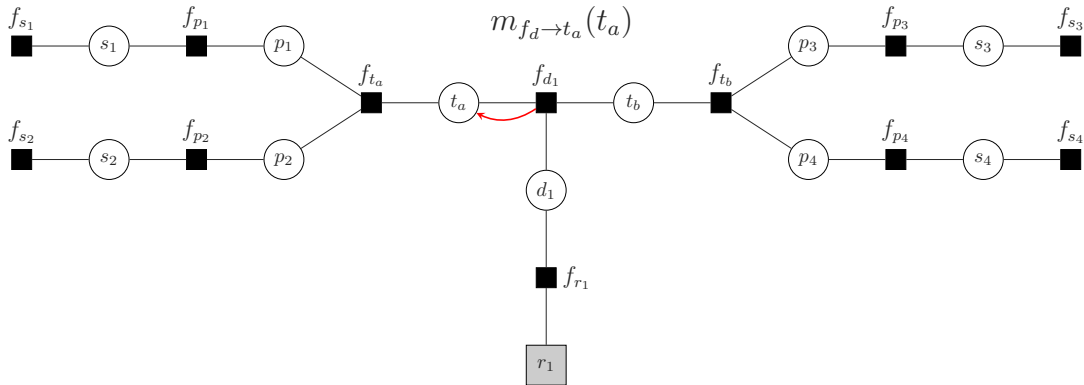
Mensajes ascendentes



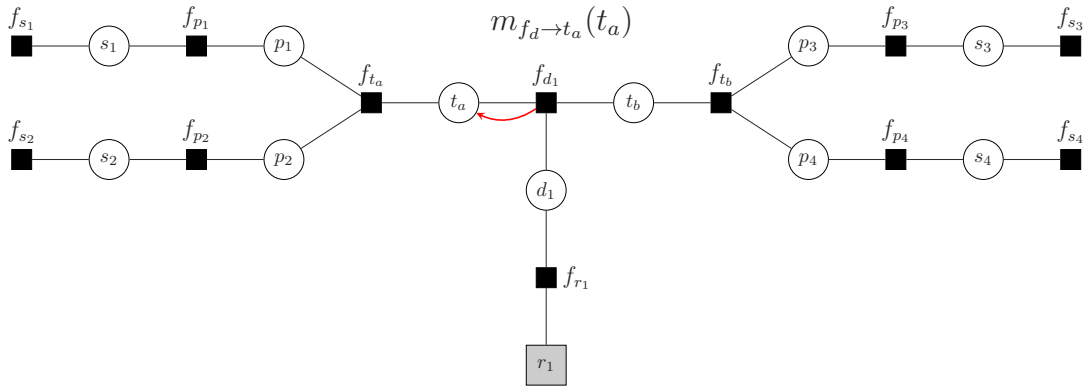
$$m_{f_r \rightarrow d}(d) =$$



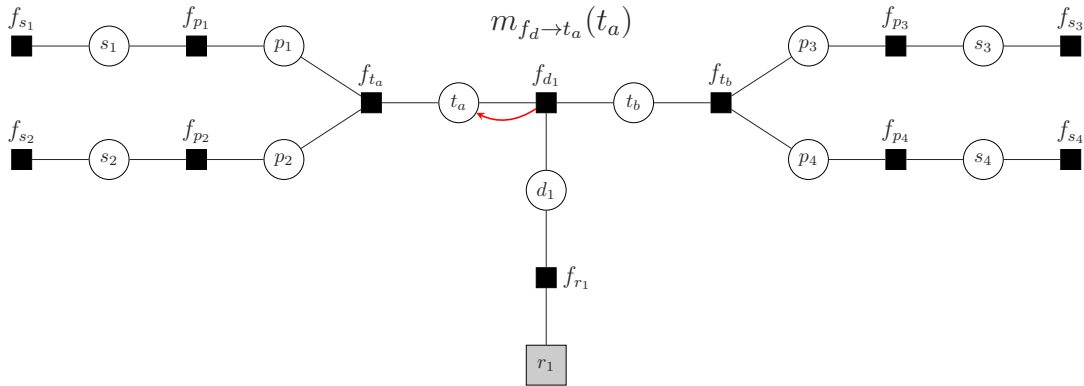
$$m_{f_r \rightarrow d}(d) = \mathbb{I}(d > 0)$$



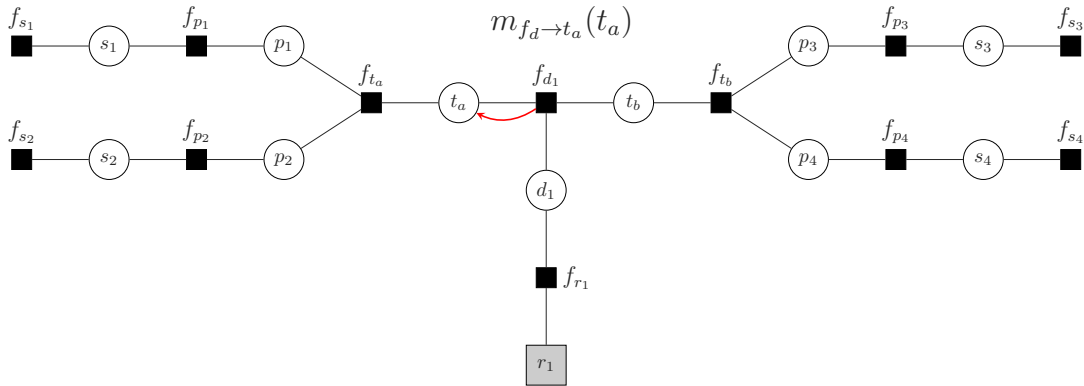
$$m_{f_d \rightarrow t_a}(t_a) =$$



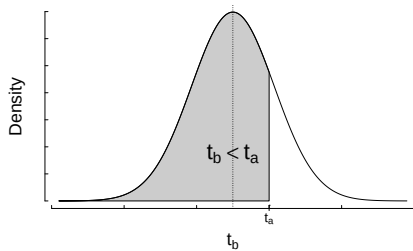
$$m_{f_d \rightarrow t_a}(t_a) = \iint \mathbb{I}(d = t_a - t_b) \mathbb{I}(d > 0) \mathcal{N}(t_b | \mu_b, \sigma_b^2) dt_b dd$$

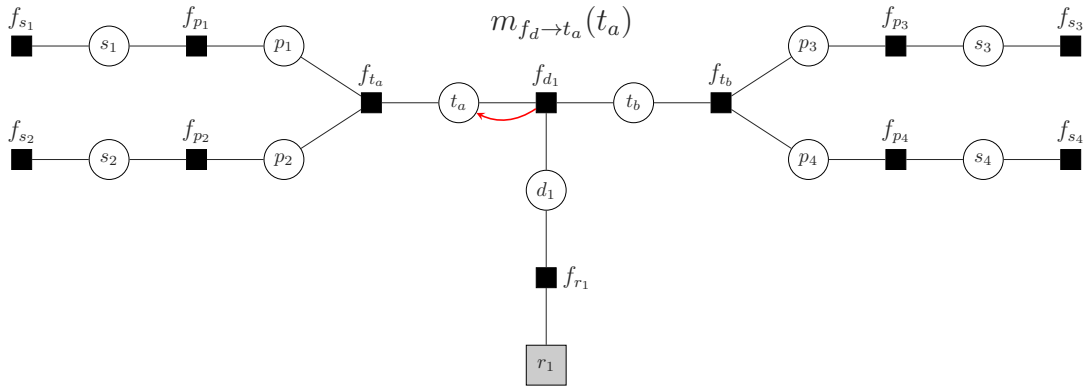


$$m_{f_d \rightarrow t_a}(t_a) = \int \mathbb{I}(t_a > t_b) \mathcal{N}(t_b | \mu_b, \sigma_b^2) dt_b$$



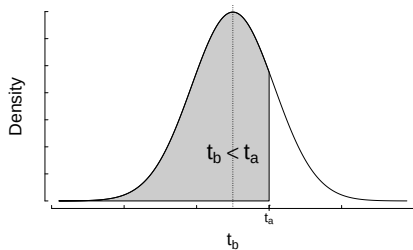
$$m_{f_d \rightarrow t_a}(t_a) = \int \mathbb{I}(t_a > t_b) \mathcal{N}(t_b | \mu_b, \sigma_b^2) dt_b$$



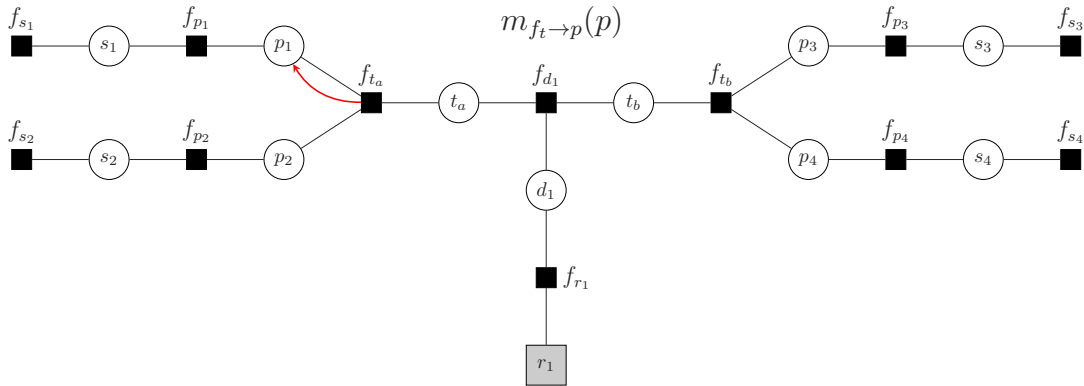


$$m_{f_d \rightarrow t_a}(t_a) = \int \mathbb{I}(t_a > t_b) \mathcal{N}(t_b | \mu_b, \sigma_b^2) dt_b$$

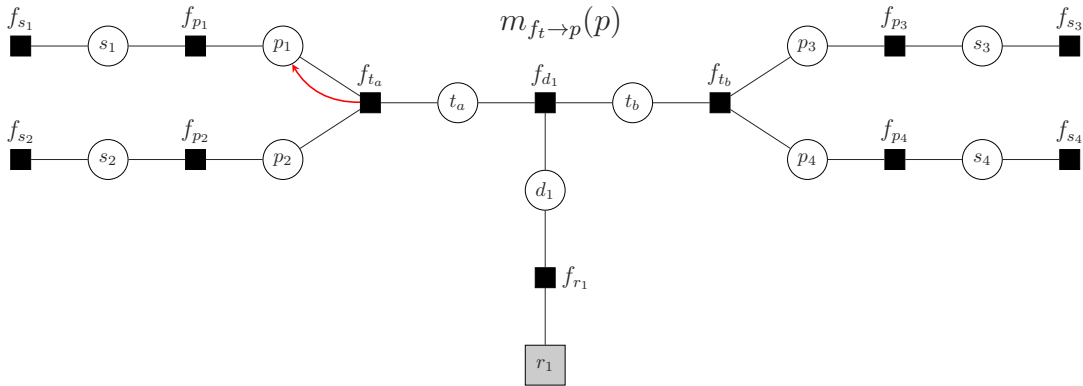
$$= \Phi(t_a | \mu_b, \sigma_b^2)$$



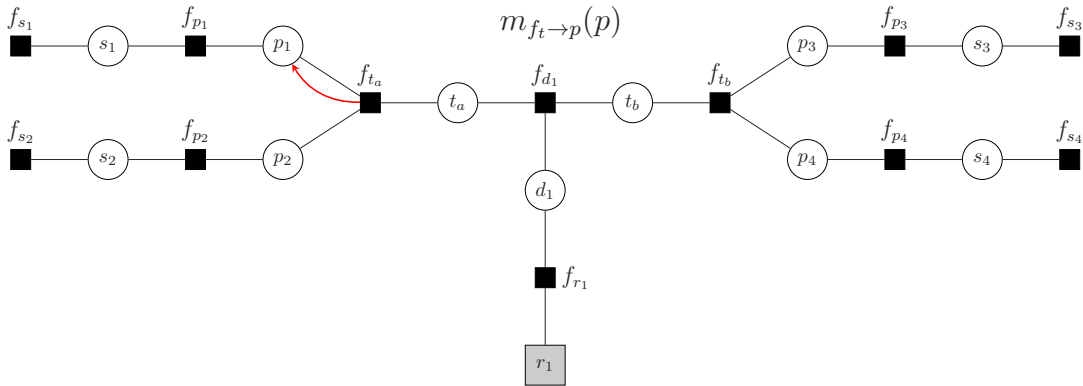




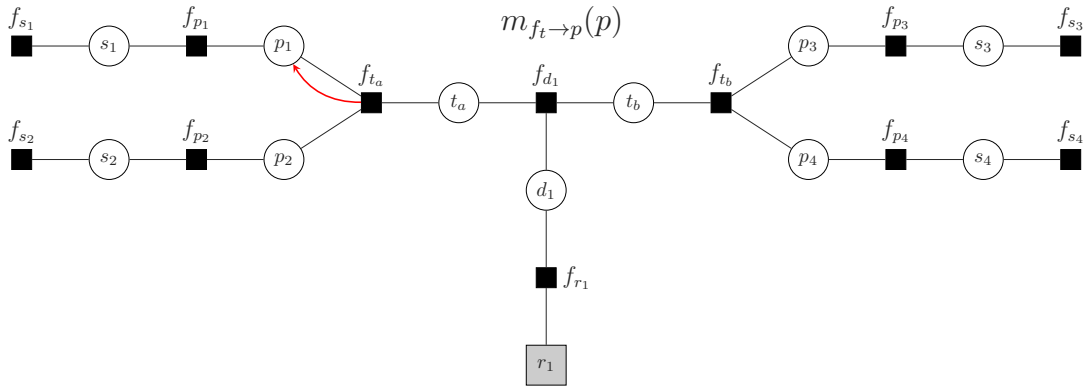
$$m_{f_{t_a} \rightarrow p_1}(p_1) =$$



$$m_{f_{t_a} \rightarrow p_1}(p_1) = \iint \mathbb{I}(t_a = p_1 + p_2) \mathcal{N}(p_2 | \mu_2, \sigma_2^2 + \beta^2) \Phi(t_a | \mu_b, \sigma_b^2) dt_a dp_2$$

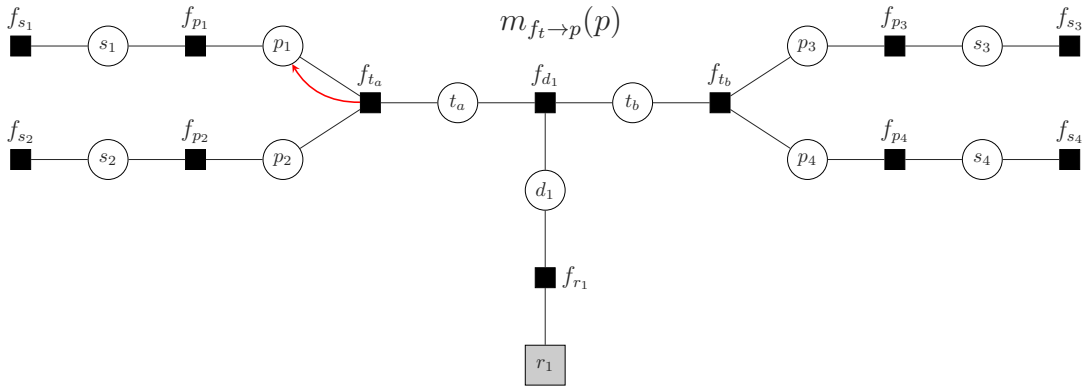


$$m_{f_{t_a} \rightarrow p_1}(p_1) = \int \mathcal{N}(p_2 | \mu_2, \sigma_2^2 + \beta^2) \Phi(p_1 + p_2 | \mu_b, \sigma_b^2) dp_2$$



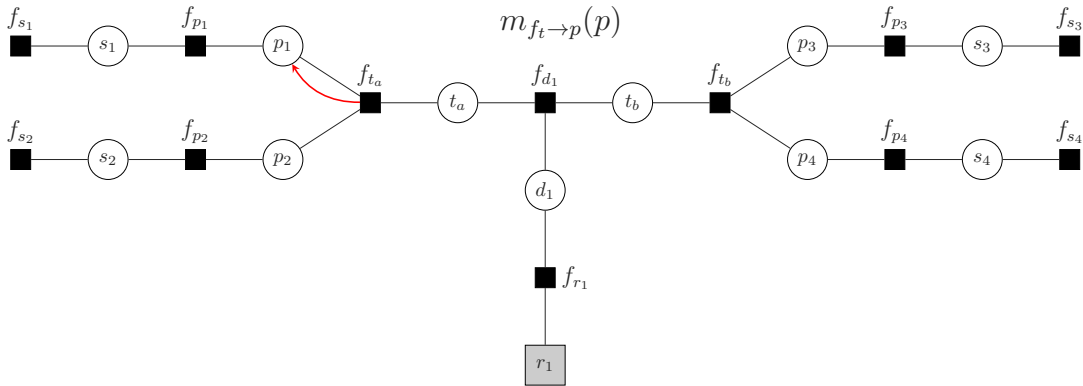
$$m_{f_t \rightarrow p}(p)$$

$$m_{f_{t_a} \rightarrow p_1}(p_1) = \int \mathcal{N}(p_2 | \mu_2, \sigma_2^2 + \beta^2) \Phi(p_1 | \mu_b - p_2, \sigma_b^2) dp_2$$



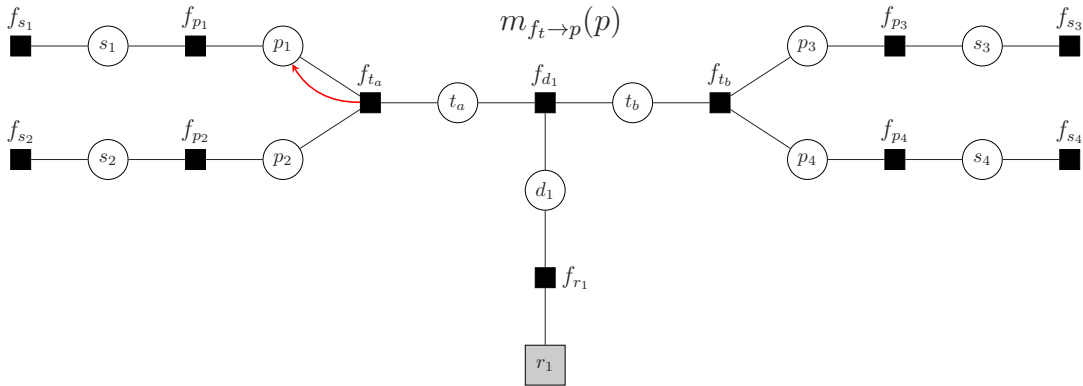
$$m_{f_{t_a} \rightarrow p_1}(p_1) = \int \mathcal{N}(p_2 | \mu_2, \sigma_2^2 + \beta^2) \Phi(p_1 | \mu_b - p_2, \sigma_b^2) dp_2$$

$$\frac{\partial}{\partial p_1} m_{f_{t_a} \rightarrow p_1}(p_1) = \frac{\partial}{\partial p_1} \int \mathcal{N}(p_2 | \mu_2, \sigma_2^2 + \beta^2) \Phi(p_1 | \mu_b - p_2, \sigma_b^2) dp_2$$



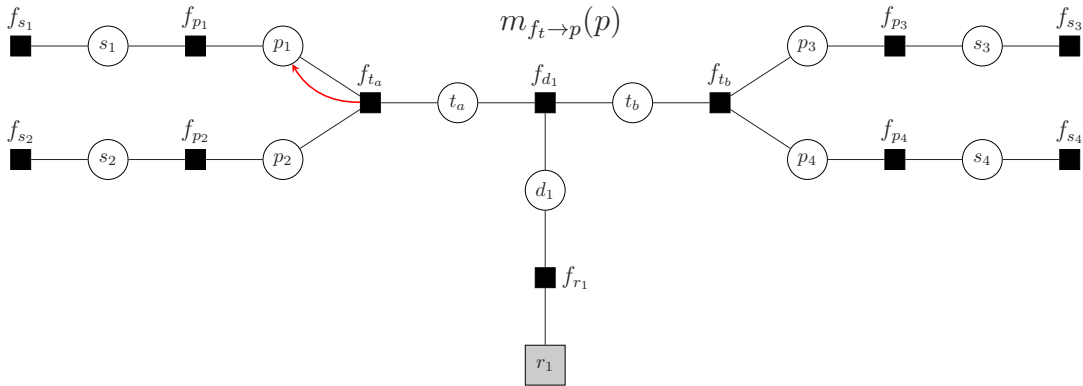
$$m_{f_{t_a} \rightarrow p_1}(p_1) = \int \mathcal{N}(p_2 | \mu_2, \sigma_2^2 + \beta^2) \Phi(p_1 | \mu_b - p_2, \sigma_b^2) dp_2$$

$$\frac{\partial}{\partial p_1} m_{f_{t_a} \rightarrow p_1}(p_1) = \int \mathcal{N}(p_2 | \mu_2, \sigma_2^2 + \beta^2) \frac{\partial}{\partial p_1} \Phi(p_1 | \mu_b - p_2, \sigma_b^2) dp_2$$



$$m_{f_{t_a} \rightarrow p_1}(p_1) = \int \mathcal{N}(p_2 | \mu_2, \sigma_2^2 + \beta^2) \Phi(p_1 | \mu_b - p_2, \sigma_b^2) dp_2$$

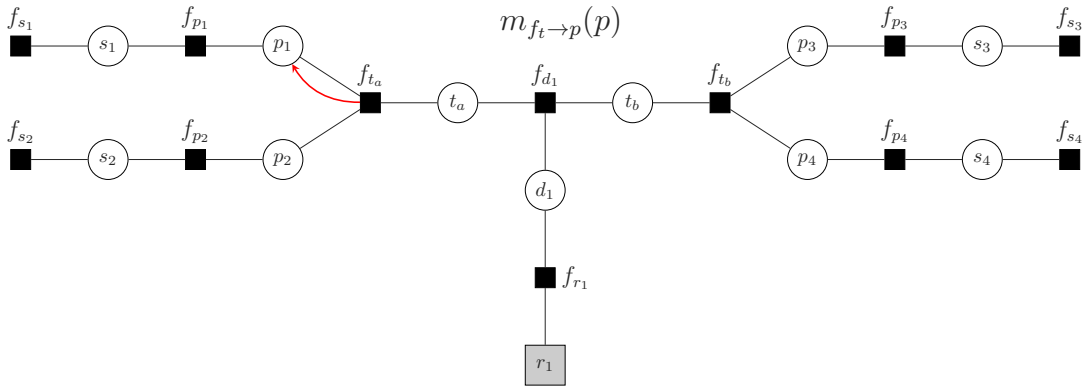
$$\frac{\partial}{\partial p_1} m_{f_{t_a} \rightarrow p_1}(p_1) = \int \mathcal{N}(p_2 | \mu_2, \sigma_2^2 + \beta^2) \mathcal{N}(p_1 | \mu_b - p_2, \sigma_b^2) dp_2$$



$$m_{f_{t_a} \rightarrow p_1}(p_1) = \int \mathcal{N}(p_2 | \mu_2, \sigma_2^2 + \beta^2) \Phi(p_1 | \mu_b - p_2, \sigma_b^2) dp_2$$

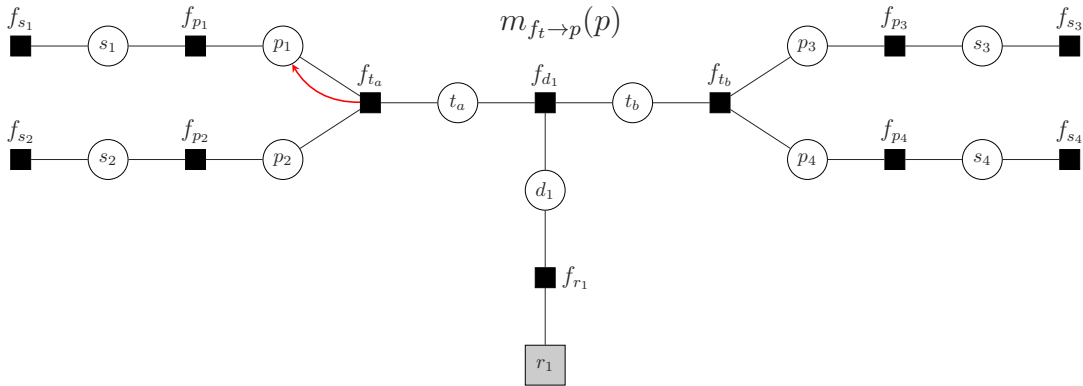
$$\frac{\partial}{\partial p_1} m_{f_{t_a} \rightarrow p_1}(p_1) = \int \mathcal{N}(p_2 | \mu_2, \sigma_2^2 + \beta^2) \mathcal{N}(p_2 | \mu_b - p_1, \sigma_b^2) dp_2$$





$$m_{f_{t_a} \rightarrow p_1}(p_1) = \int \mathcal{N}(p_2 | \mu_2, \sigma_2^2 + \beta^2) \Phi(p_1 | \mu_b - p_2, \sigma_b^2) dp_2$$

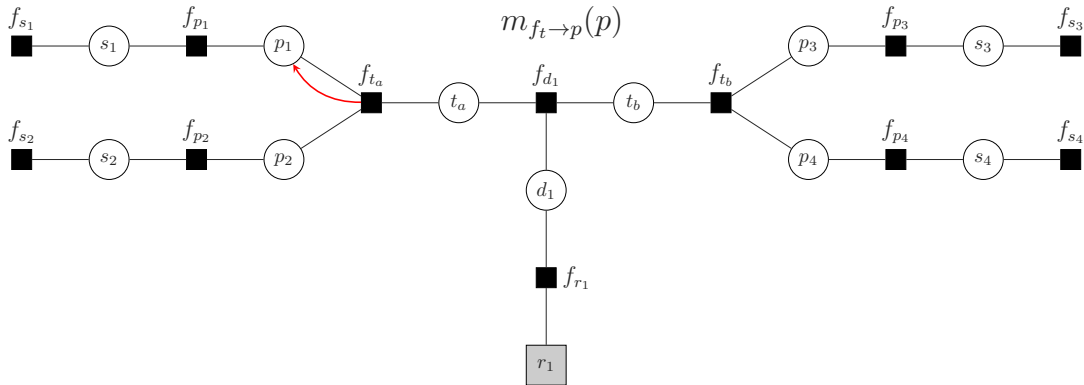
$$\frac{\partial}{\partial p_1} m_{f_{t_a} \rightarrow p_1}(p_1) = \int \underbrace{\mathcal{N}(\mu_2 | \mu_b - p_1, \sigma_b^2 + \sigma_2^2 + \beta^2)}_{\text{const.}} \underbrace{\mathcal{N}(p_2 | \mu_*, \sigma_*^2) dp_2}_1$$



$$m_{f_t \rightarrow p}(p)$$

$$m_{f_{t_a} \rightarrow p_1}(p_1) = \int \mathcal{N}(p_2 | \mu_2, \sigma_2^2 + \beta^2) \Phi(p_1 | \mu_b - p_2, \sigma_b^2) dp_2$$

$$\frac{\partial}{\partial p_1} m_{f_{t_a} \rightarrow p_1}(p_1) = \mathcal{N}(p_1 | \mu_b - \mu_2, \sigma_b^2 + \sigma_2^2 + \beta^2)$$

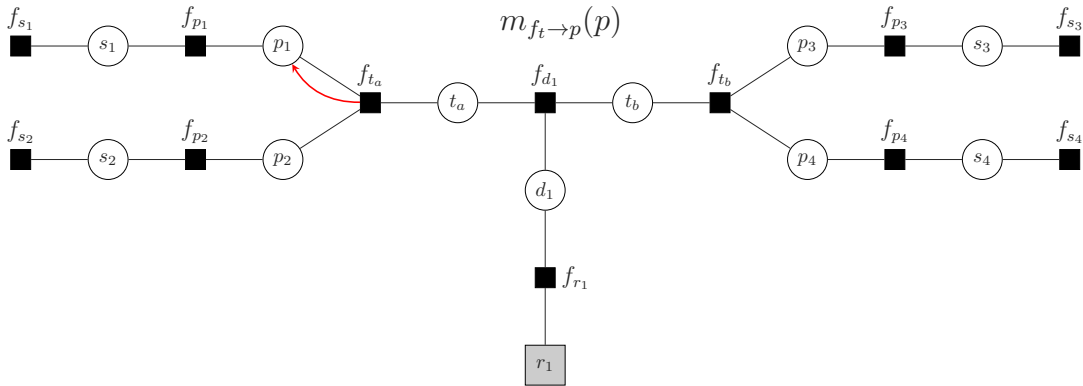


$$m_{f_t \rightarrow p}(p)$$

$$m_{f_{t_a} \rightarrow p_1}(p_1) = \int \mathcal{N}(p_2 | \mu_2, \sigma_2^2 + \beta^2) \Phi(p_1 | \mu_b - p_2, \sigma_b^2) dp_2$$

$$\frac{\partial}{\partial p_1} m_{f_{t_a} \rightarrow p_1}(p_1) = \mathcal{N}(p_1 | \mu_b - \mu_2, \sigma_b^2 + \sigma_2^2 + \beta^2)$$

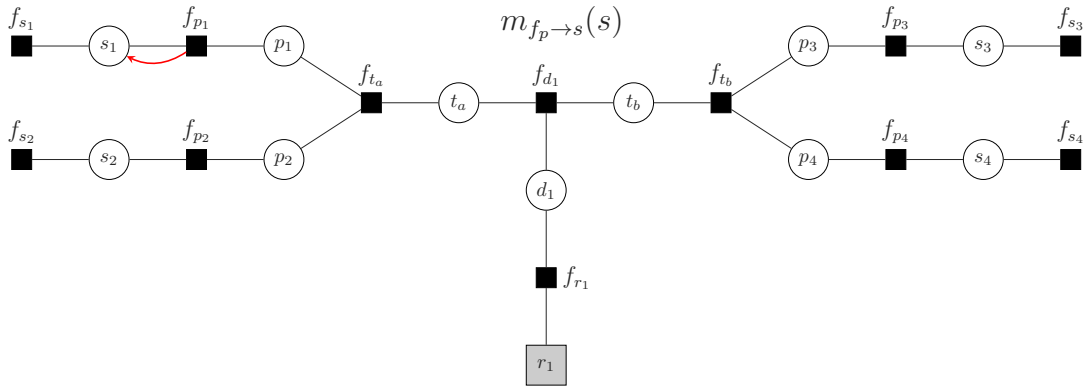
$$m_{f_{t_a} \rightarrow p_1}(p_1) =$$



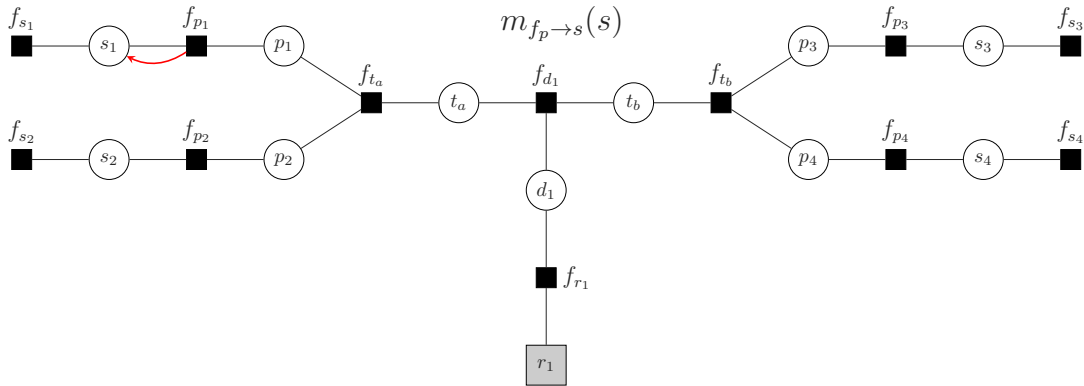
$$m_{f_{t_a} \rightarrow p_1}(p_1) = \int \mathcal{N}(p_2 | \mu_2, \sigma_2^2 + \beta^2) \Phi(p_1 | \mu_b - p_2, \sigma_b^2) dp_2$$

$$\frac{\partial}{\partial p_1} m_{f_{t_a} \rightarrow p_1}(p_1) = \mathcal{N}(p_1 | \mu_b - \mu_2, \sigma_b^2 + \sigma_2^2 + \beta^2)$$

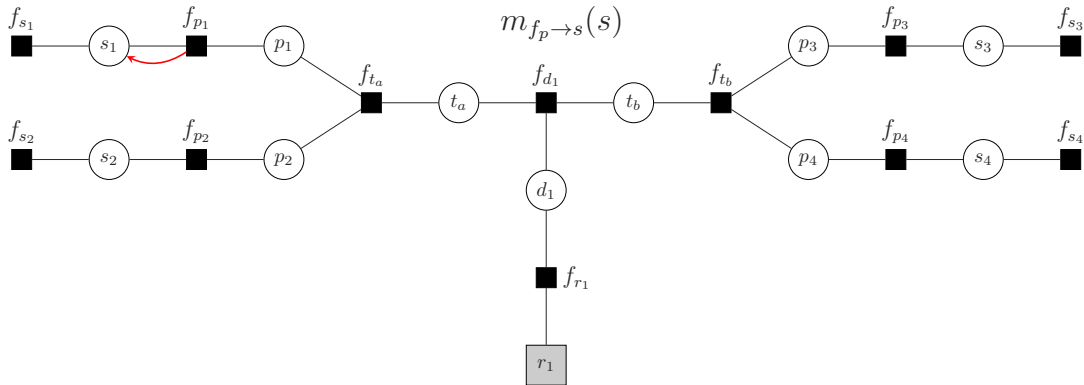
$$m_{f_{t_a} \rightarrow p_1}(p_1) = \Phi(p_1 | \mu_b - \mu_2, \sigma_b^2 + \sigma_2^2 + \beta^2)$$



$$m_{f_{p_1} \rightarrow s_1}(s_1) =$$

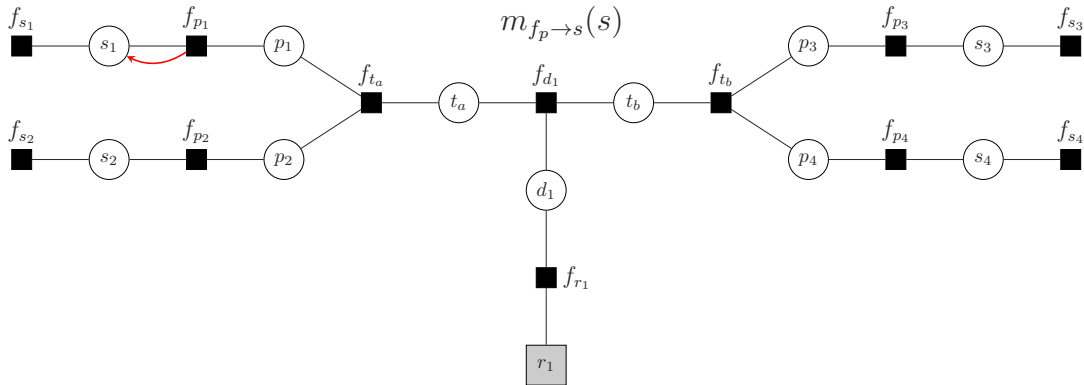


$$m_{f_{p_1} \rightarrow s_1}(s_1) = \int \mathcal{N}(p_1 | s_1, \beta^2) \Phi(p_1 | \mu_b - \mu_2, \sigma_b^2 + \sigma_2^2 + \beta^2) dp_1$$



$$m_{f_{p_1} \rightarrow s_1}(s_1) = \int \mathcal{N}(p_1 | s_1, \beta^2) \Phi(p_1 | \mu_b - \mu_2, \sigma_b^2 + \sigma_2^2 + \beta^2) dp_1$$

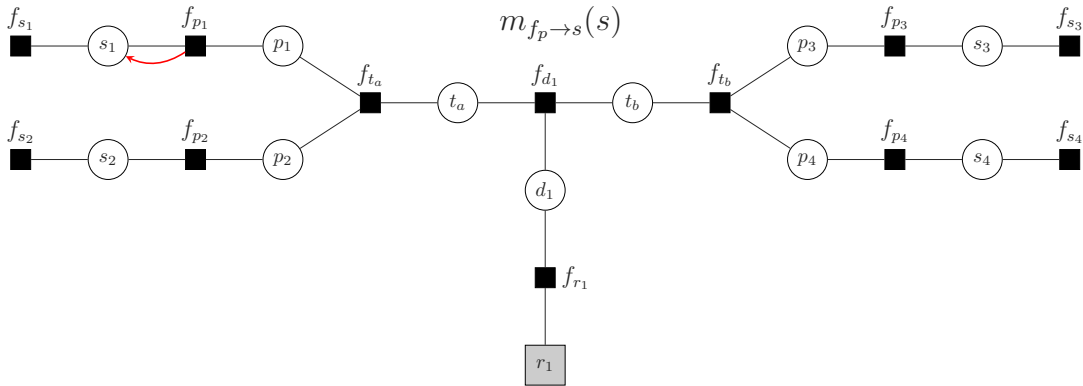
¿Sobre cuál variable derivamos?



$$m_{f_{p_1} \rightarrow s_1}(s_1) = \int \mathcal{N}(p_1 | s_1, \beta^2) \Phi(\mu_2 | \mu_b - p_1, \sigma_b^2 + \sigma_2^2 + \beta^2) dp_1$$

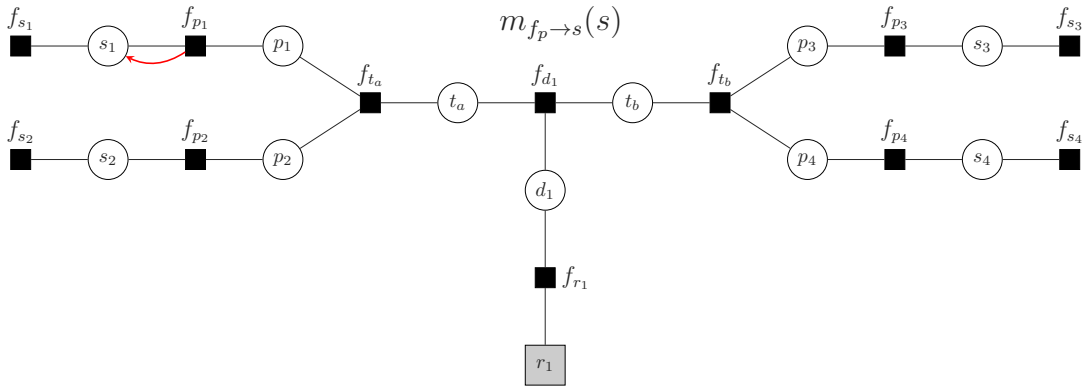
¿Sobre cuál variable derivamos?





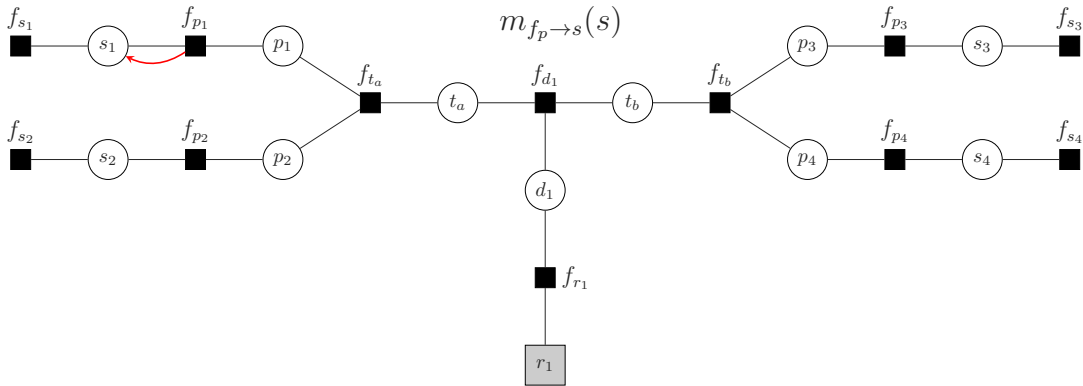
$$m_{f_{p_1} \rightarrow s_1}(s_1) = \int \mathcal{N}(p_1 | s_1, \beta^2) \Phi(\mu_2 | \mu_b - p_1, \sigma_b^2 + \sigma_2^2 + \beta^2) dp_1$$

$$\frac{\partial}{\partial \mu_2} m_{f_{s_1} \rightarrow s_1}(s_1) =$$



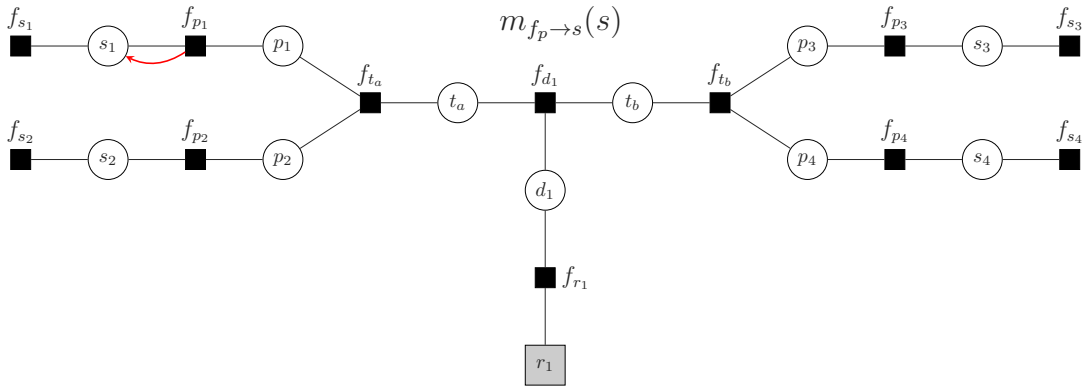
$$m_{f_{p_1} \rightarrow s_1}(s_1) = \int \mathcal{N}(p_1 | s_1, \beta^2) \Phi(\mu_2 | \mu_b - p_1, \sigma_b^2 + \sigma_2^2 + \beta^2) dp_1$$

$$\frac{\partial}{\partial \mu_2} m_{f_{s_1} \rightarrow s_1}(s_1) = \int \mathcal{N}(p_1 | s_1, \beta^2) \mathcal{N}(p_1 | \mu_b - \mu_2, \sigma_b^2 + \sigma_2^2 + \beta^2) dp_1$$



$$m_{f_{p_1} \rightarrow s_1}(s_1) = \int \mathcal{N}(p_1 | s_1, \beta^2) \Phi(\mu_2 | \mu_b - p_1, \sigma_b^2 + \sigma_2^2 + \beta^2) dp_1$$

$$\frac{\partial}{\partial \mu_2} m_{f_{s_1} \rightarrow s_1}(s_1) = \mathcal{N}(s_1 | \mu_b - \mu_2, \sigma_b^2 + \sigma_2^2 + 2\beta^2)$$

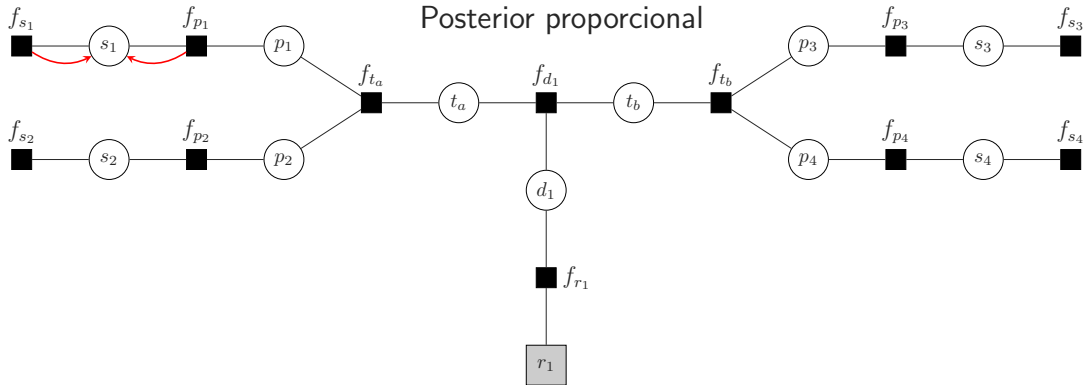


$$m_{f_{p_1} \rightarrow s_1}(s_1) = \int \mathcal{N}(p_1 | s_1, \beta^2) \Phi(\mu_2 | \mu_b - p_1, \sigma_b^2 + \sigma_2^2 + \beta^2) dp_1$$

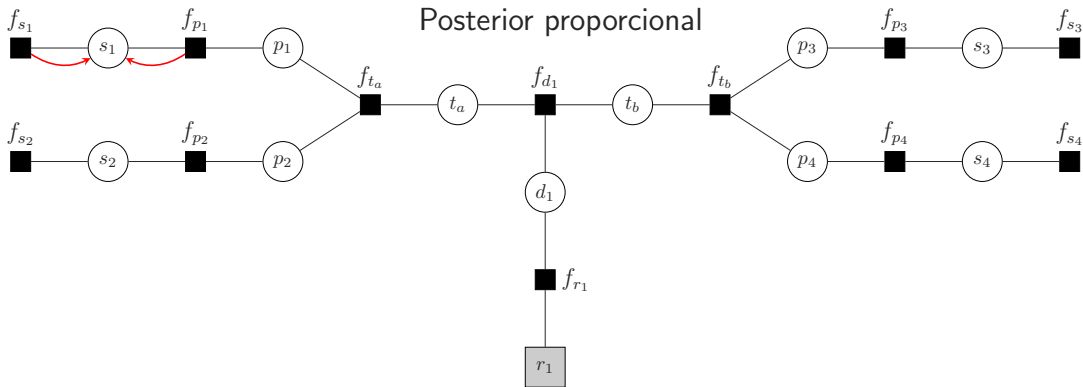
$$\frac{\partial}{\partial \mu_2} m_{f_{s_1} \rightarrow s_1}(s_1) = \mathcal{N}(s_1 | \mu_b - \mu_2, \sigma_b^2 + \sigma_2^2 + 2\beta^2)$$

$$m_{f_{p_1} \rightarrow s_1}(s_1) = \Phi(s_1 | \mu_b - \mu_2, \sigma_b^2 + \sigma_2^2 + 2\beta^2)$$

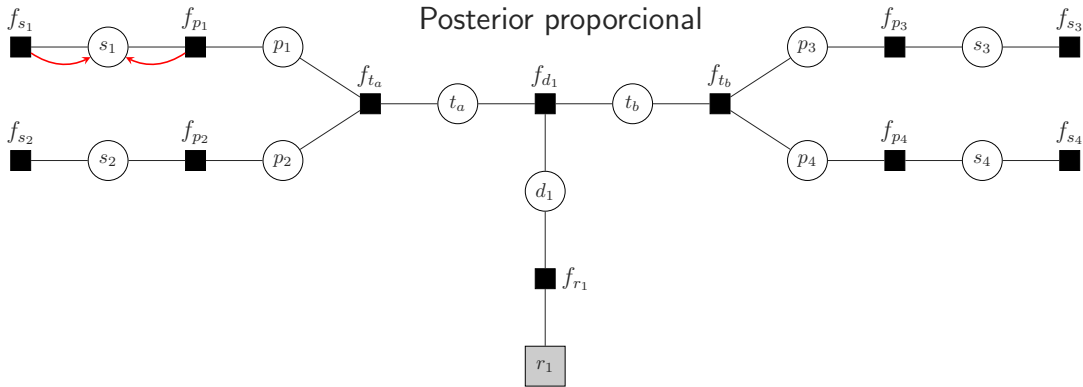
# Posterior proporcional



$$p(s_1|r) \propto p(s_1, r) =$$



$$p(s_1|r) \propto p(s_1, r) = \mathcal{N}(s_1|\mu_1, \sigma_1^2) \Phi(s_1|\mu_b - \mu_2, \sigma_b^2 + \sigma_2^2 + 2\beta^2)$$



$$p(s_1|r) \propto p(s_1, r) = \underbrace{\mathcal{N}(s_1|\mu_1, \sigma_1^2)}_{\text{Prior}} \underbrace{\Phi(s_1|\mu_b - \mu_2, \sigma_b^2 + \sigma_2^2 + 2\beta^2)}_{\text{Verosimilitud}}$$

# Verosimilitud

$$p(r|s_1) = \Phi(s_1|\mu_b - \mu_2, \sigma_b^2 + \sigma_2^2 + 2\beta^2)$$



# Verosimilitud

$$p(r|s_1) = \Phi(s_1|\mu_b - \mu_2, \sigma_b^2 + \sigma_2^2 + 2\beta^2)$$

$$= \Phi(0|\mu_b - \mu_2 - s_1, \sigma_b^2 + \sigma_2^2 + 2\beta^2)$$

# Verosimilitud

$$\begin{aligned} p(r|s_1) &= \Phi(s_1|\mu_b - \mu_2, \sigma_b^2 + \sigma_2^2 + 2\beta^2) \\ &= \Phi(0|\boldsymbol{\mu}_b - \mu_2 - s_1, \boldsymbol{\sigma}_b^2 + \sigma_2^2 + 2\beta^2) \end{aligned}$$

# Verosimilitud

$$p(r|s_1) = \Phi(s_1|\mu_b - \mu_2, \sigma_b^2 + \sigma_2^2 + 2\beta^2)$$

$$= \Phi(0|\boldsymbol{\mu_3} + \boldsymbol{\mu_4} - \mu_2 - s_1, \boldsymbol{\sigma_3^2} + \boldsymbol{\sigma_4^2} + \boldsymbol{2\beta^2} + \sigma_2^2 + 2\beta^2)$$

# Verosimilitud

$$\begin{aligned} p(r|s_1) &= \Phi(s_1|\mu_b - \mu_2, \sigma_b^2 + \sigma_2^2 + 2\beta^2) \\ &= \Phi(0 | (\mu_3 + \mu_4) - (s_1 + \mu_2), 4\beta^2 + \sigma_2^2 + \sigma_3^2 + \sigma_4^2) \\ &= \end{aligned}$$

# Verosimilitud

$$\begin{aligned}p(r|s_1) &= \Phi(s_1|\mu_b - \mu_2, \sigma_b^2 + \sigma_2^2 + 2\beta^2) \\&= \Phi(0 | (\mu_3 + \mu_4) - (s_1 + \mu_2), 4\beta^2 + \sigma_2^2 + \sigma_3^2 + \sigma_4^2) \\&= 1 - \Phi(0 | (s_1 + \mu_2) - (\mu_3 + \mu_4), 4\beta^2 + \sigma_2^2 + \sigma_3^2 + \sigma_4^2)\end{aligned}$$

# Verosimilitud

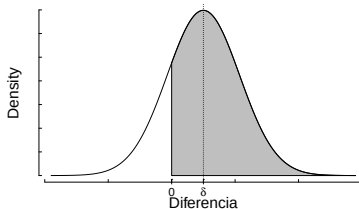
$$\begin{aligned} p(r|s_1) &= \Phi(s_1|\mu_b - \mu_2, \sigma_b^2 + \sigma_2^2 + 2\beta^2) \\ &= \Phi(0|(\mu_3 + \mu_4) - (s_1 + \mu_2), 4\beta^2 + \sigma_2^2 + \sigma_3^2 + \sigma_4^2) \\ &= 1 - \Phi(0|\underbrace{(s_1 + \mu_2) - (\mu_3 + \mu_4)}_{\substack{\text{Diferencia esperada} \\ \text{si la habilidad fuera } s_1}}, \underbrace{4\beta^2 + \sigma_2^2 + \sigma_3^2 + \sigma_4^2}_{\substack{\text{Incertidumbre total} \\ \text{salvo la de la habilidad } s_1}}) \end{aligned}$$

# Evidencia vs Verosimilitud

$$p(r) = 1 - \Phi\left(0 \mid \underbrace{(\mu_1 + \mu_2) - (\mu_3 + \mu_4)}_{\text{Diferencia esperada}}, \underbrace{4\beta^2 + \sigma_1^2 + \sigma_2^2 + \sigma_3^2 + \sigma_4^2}_{\text{Incertidumbre total}}\right)$$

$$p(r|s_1) = 1 - \Phi\left(0 \mid \underbrace{(s_1 + \mu_2) - (\mu_3 + \mu_4)}_{\text{Diferencia esperada si la habilidad fuera } s_1}, \underbrace{4\beta^2 + \sigma_2^2 + \sigma_3^2 + \sigma_4^2}_{\text{Incertidumbre total salvo la de la habilidad } s_1}\right)$$

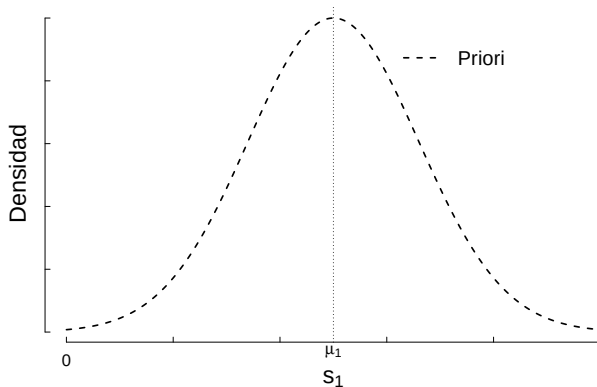
Predicciones del resultado  
usando todas las hipótesis  
pesadas por la creencia a prior



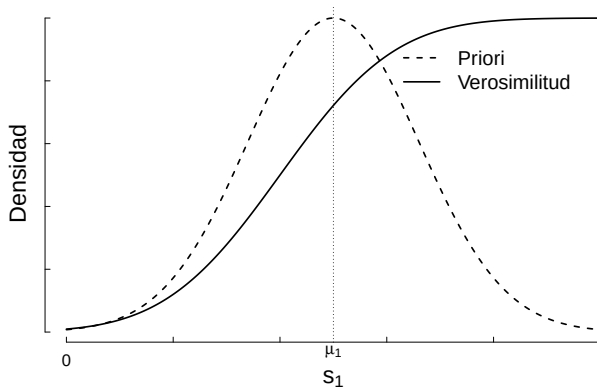
$$\overbrace{P(s_1 \mid r, \text{Modelo})}^{\text{Posterior}} \propto \overbrace{\mathcal{N}(s_1 \mid \mu_1, \sigma_1^2)}^{\text{Prior}} \overbrace{1 - \Phi(0 \mid \delta - \mu_1 + s_1, \vartheta^2 - \sigma_1^2)}^{\text{Verosimilitud}} \quad \text{Caso ganador}$$



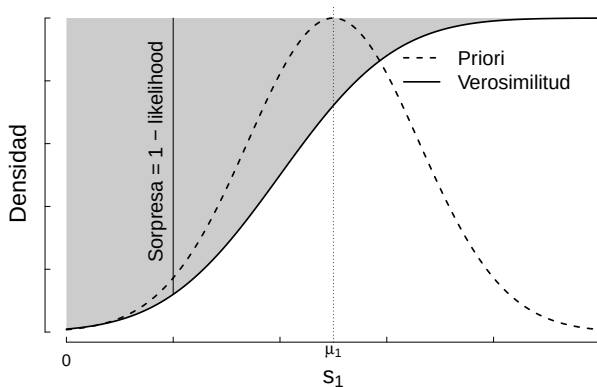
$$\overbrace{P(s_1 \mid r, \text{Modelo})}^{\text{Posterior}} \propto \overbrace{\mathcal{N}(s_1 \mid \mu_1, \sigma_1^2)}^{\text{Prior}} \overbrace{1 - \Phi(0 \mid \delta - \mu_1 + s_1, \vartheta^2 - \sigma_1^2)}^{\text{Verosimilitud}} \quad \text{Caso ganador}$$



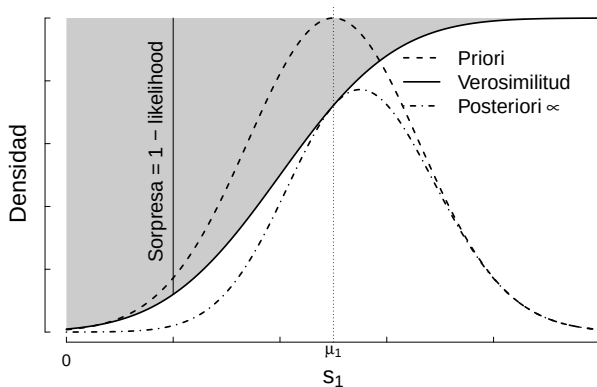
$$\overbrace{P(s_1 \mid r, \text{Modelo})}^{\text{Posterior}} \propto \overbrace{\mathcal{N}(s_1 \mid \mu_1, \sigma_1^2)}^{\text{Prior}} \overbrace{1 - \Phi(0 \mid \delta - \mu_1 + s_1, \vartheta^2 - \sigma_1^2)}^{\text{Verosimilitud}} \quad \text{Caso ganador}$$



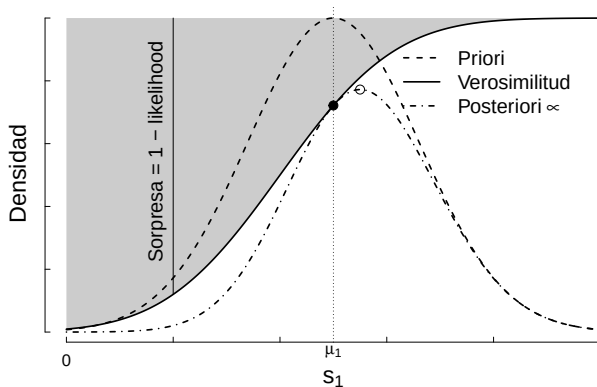
$$\overbrace{P(s_1 \mid r, \text{Modelo})}^{\text{Posterior}} \propto \overbrace{\mathcal{N}(s_1 \mid \mu_1, \sigma_1^2)}^{\text{Prior}} \overbrace{1 - \Phi(0 \mid \delta - \mu_1 + s_1, \vartheta^2 - \sigma_1^2)}^{\text{Verosimilitud}} \quad \text{Caso ganador}$$



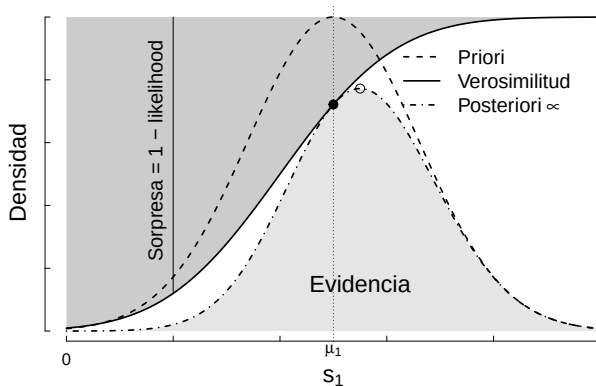
$$\overbrace{P(s_1 \mid r, \text{Modelo})}^{\text{Posterior}} \propto \overbrace{\mathcal{N}(s_1 \mid \mu_1, \sigma_1^2)}^{\text{Prior}} \overbrace{1 - \Phi(0 \mid \delta - \mu_1 + s_1, \vartheta^2 - \sigma_1^2)}^{\text{Verosimilitud}} \quad \text{Caso ganador}$$



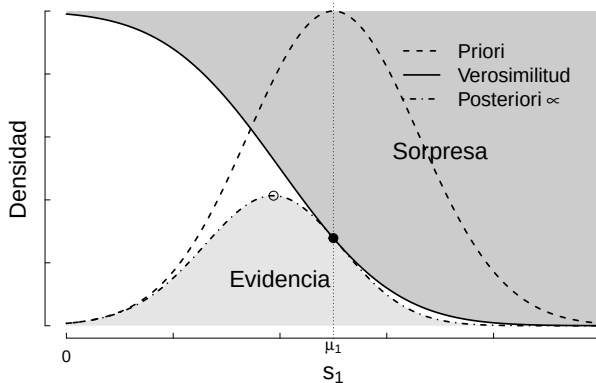
$$\overbrace{P(s_1 \mid r, \text{Modelo})}^{\text{Posterior}} \propto \overbrace{\mathcal{N}(s_1 \mid \mu_1, \sigma_1^2)}^{\text{Prior}} \overbrace{1 - \Phi(0 \mid \delta - \mu_1 + s_1, \vartheta^2 - \sigma_1^2)}^{\text{Verosimilitud}} \quad \text{Caso ganador}$$



$$\overbrace{P(s_1 \mid r, \text{Modelo})}^{\text{Posterior}} \propto \overbrace{\mathcal{N}(s_1 \mid \mu_1, \sigma_1^2)}^{\text{Prior}} \overbrace{1 - \Phi(0 \mid \delta - \mu_1 + s_1, \vartheta^2 - \sigma_1^2)}^{\text{Verosimilitud}} \quad \text{Caso ganador}$$



$$\overbrace{P(s_3 \mid r, \text{Modelo})}^{\text{Posteriori}} \propto \overbrace{N(s_3 \mid \mu_3, \sigma_2^2)}^{\text{Priori}} \overbrace{\Phi(0 \mid \delta - \mu_3 - s_3, \vartheta^2 - \sigma_3^2)}^{\text{Verosimilitud}} \quad \text{Caso perdedor}$$



# TrueSkill

Posterior aproximado

$$\underbrace{\hat{p}(s_a | \text{Gana}, M)}_{\text{Aproximado}} = \arg \min_{\mu, \sigma} \mathcal{D}_{KL} \left( \underbrace{p(s_a | \text{Gana}, M)}_{\text{Exacto}} \parallel \underbrace{\mathcal{N}(s_a | \mu, \sigma^2)}_{\text{Familia}} \right)$$



# Mínima Divergencia

Posterior aproximado

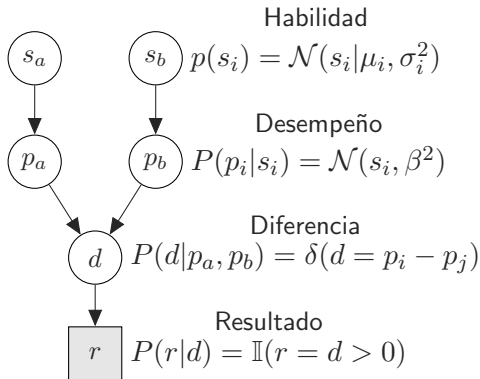
$$\underbrace{\hat{p}(s_a|\text{Gana}, M)}_{\text{Aproximado}} = \arg \min_{\mu, \sigma} \mathcal{D}_{KL} \left( \underbrace{p(s_a|\text{Gana}, M)}_{\text{Exacto}} \parallel \underbrace{\mathcal{N}(s_a|\mu, \sigma^2)}_{\text{Familia}} \right)$$

$$\mathcal{D}_{KL}(\text{Exacta}||\text{Familia}) = \sum_{s_a} p(s_a|\text{Gana}, M) \cdot \left( \log p(s_a|\text{Gana}, M) - \log \mathcal{N}(s_a|\mu, \sigma^2) \right)$$

# Mínima Divergencia

Posterior aproximado

$$\underbrace{\hat{p}(s_a|\text{Gana}, M)}_{\text{Aproximado}} = \arg \min_{\mu, \sigma} \mathcal{D}_{KL} \left( \underbrace{p(s_a|\text{Gana}, M)}_{\text{Exacto}} \parallel \underbrace{\mathcal{N}(s_a|\mu, \sigma^2)}_{\text{Familia}} \right)$$



# Mínima Divergencia

Posterior aproximado

$$\underbrace{\hat{p}(s_a | \text{Gana}, M)}_{\text{Aproximado}} = \arg \min_{\mu, \sigma} \mathcal{D}_{KL} \left( \underbrace{p(s_a | \text{Gana}, M)}_{\text{Exacto}} \parallel \underbrace{\mathcal{N}(s_a | \mu, \sigma^2)}_{\text{Familia}} \right)$$

$$p(d, \text{Gana} \mid \text{Modelo}) =$$

$$\underbrace{\hspace{10em}}_{\text{Marginal exacta}}$$

# Mínima Divergencia

Posterior aproximado

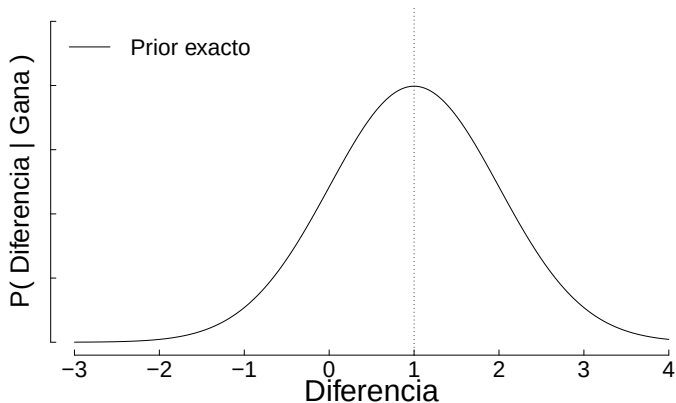
$$\underbrace{\hat{p}(s_a | \text{Gana}, M)}_{\text{Aproximado}} = \arg \min_{\mu, \sigma} \mathcal{D}_{KL} \left( \underbrace{p(s_a | \text{Gana}, M)}_{\text{Exacto}} \parallel \underbrace{\mathcal{N}(s_a | \mu, \sigma^2)}_{\text{Familia}} \right)$$

$$\underbrace{p(d, \text{Gana} \mid \text{Modelo})}_{\text{Marginal exacta}} = \underbrace{\int_{s_a} \int_{p_a} \int_{s_b} \int_{p_b} p(s_a) p(s_b) p(p_a | s_a) p(p_b | s_b) p(d | p_a, p_b) P(\text{Gana} | d)}_{\text{Prior exacto } p(d) = \mathcal{N}(d \mid \underbrace{\mu_a - \mu_b}_{\text{Diferencia de medias}}, \underbrace{\sigma_a^2 + \sigma_b^2 + 2\beta^2}_{\text{Suma de incertidumbres})}}$$

# Mínima Divergencia

Posterior aproximado

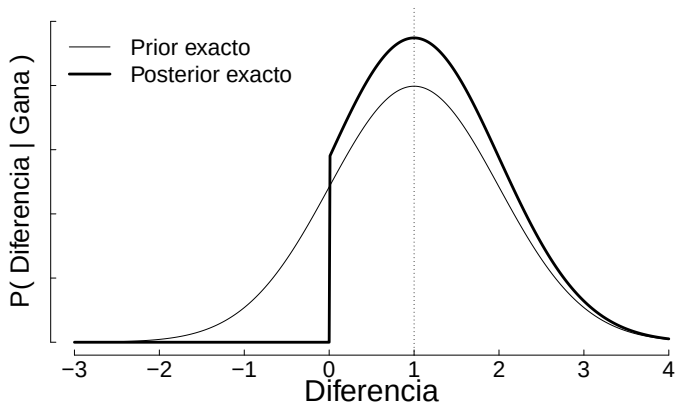
$$p(d, \text{Gana}) = \mathcal{N}(d | \mu_a - \mu_b, \sigma_a^2 + \sigma_b^2 + 2\beta^2) \mathbb{I}(d > 0)$$



# Mínima Divergencia

Posterior aproximado

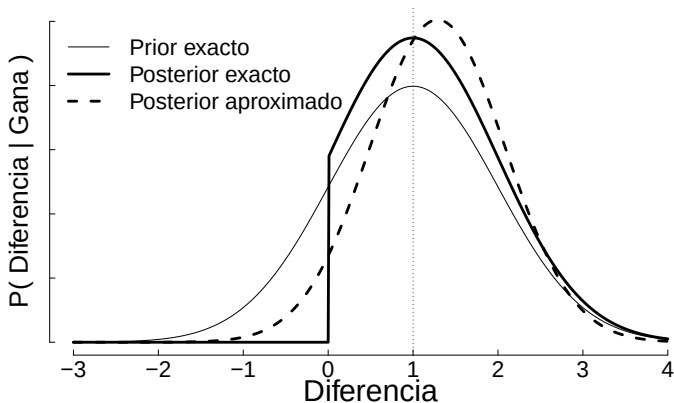
$$p(d, \text{Gana}) = \mathcal{N}(d | \mu_a - \mu_b, \sigma_a^2 + \sigma_b^2 + 2\beta^2) \mathbb{I}(d > 0)$$



# Mínima Divergencia

Posterior aproximado

$$p(d, \text{Gana}) = \mathcal{N}(d | \mu_a - \mu_b, \sigma_a^2 + \sigma_b^2 + 2\beta^2) \mathbb{I}(d > 0)$$



# Mínima Divergencia

Posterior aproximado

$$p(d, \text{Gana}) = \mathcal{N}(d | \mu_a - \mu_b, \sigma_a^2 + \sigma_b^2 + 2\beta^2) \mathbb{I}(d > 0)$$

$$\hat{p}(d, \text{Gana}) \propto \mathcal{N}\left(d \mid \underbrace{E(d, \text{Gana} | d > 0)}_{\text{Misma media}}, \underbrace{V(d, \text{Gana} | d > 0)}_{\text{Misma varianza}}\right)$$



# Mínima Divergencia

Posterior aproximado

$$p(d, \text{Gana}) = \mathcal{N}(d | \mu_a - \mu_b, \sigma_a^2 + \sigma_b^2 + 2\beta^2) \mathbb{I}(d > 0)$$

$$\hat{p}(d, \text{Gana}) \propto \mathcal{N}\left(d \mid \underbrace{E(d, \text{Gana} | d > 0)}_{\text{Misma media}}, \underbrace{V(d, \text{Gana} | d > 0)}_{\text{Misma varianza}}\right)$$

La divergencia es mínima cuando tienen mismos momentos.  
(Expectation Propagation)

# Mínima Divergencia

Aproximación del posterior de la diferencia

$$p(d, \text{Gana}) = m_{f_d \rightarrow d}(d) m_{f_r \rightarrow d}(d)$$

# Mínima Divergencia

Aproximación del posterior de la diferencia

$$p(d, \text{Gana}) = m_{f_d \rightarrow d}(d) m_{f_r \rightarrow d}(d)$$

$$m_{f_r \rightarrow d}(d) = m_{d \rightarrow f_d}(d)$$

# Mínima Divergencia

Aproximación del posterior de la diferencia

$$p(d, \text{Gana}) = m_{f_d \rightarrow d}(d) m_{d \rightarrow f_d}(d)$$

# Mínima Divergencia

Aproximación del posterior de la diferencia

$$p(d, \text{Gana}) = \underbrace{m_{f_d \rightarrow d}(d)}_{P(d)} m_{d \rightarrow f_d}(d)$$

# Mínima Divergencia

Aproximación del posterior de la diferencia

$$p(d, \text{Gana}) = \underbrace{m_{f_d \rightarrow d}(d)}_{P(d)} m_{d \rightarrow f_d}(d)$$

$$m_{d \rightarrow f_d}(d) = \frac{p(d, \text{Gana})}{p(d)}$$

# Mínima Divergencia

Aproximación del posterior de la diferencia

$$p(d, \text{Gana}) = \underbrace{m_{f_d \rightarrow d}(d)}_{P(d)} m_{d \rightarrow f_d}(d)$$

$$m_{d \rightarrow f_d}(d) = \frac{p(d, \text{Gana})}{p(d)} \approx \frac{\hat{p}(d, \text{Gana})}{p(d)}$$

# Mínima Divergencia

Aproximación del posterior de la diferencia

$$p(d, \text{Gana}) = \underbrace{m_{f_d \rightarrow d}(d)}_{P(d)} m_{d \rightarrow f_d}(d)$$

$$\begin{aligned} m_{d \rightarrow f_d}(d) &= \frac{p(d, \text{Gana})}{p(d)} \approx \frac{\hat{p}(d, \text{Gana})}{p(d)} \\ &= \frac{\mathcal{N}(d | \hat{\delta}, \hat{\vartheta}^2)}{\mathcal{N}(d | \delta, \vartheta^2)} \end{aligned}$$



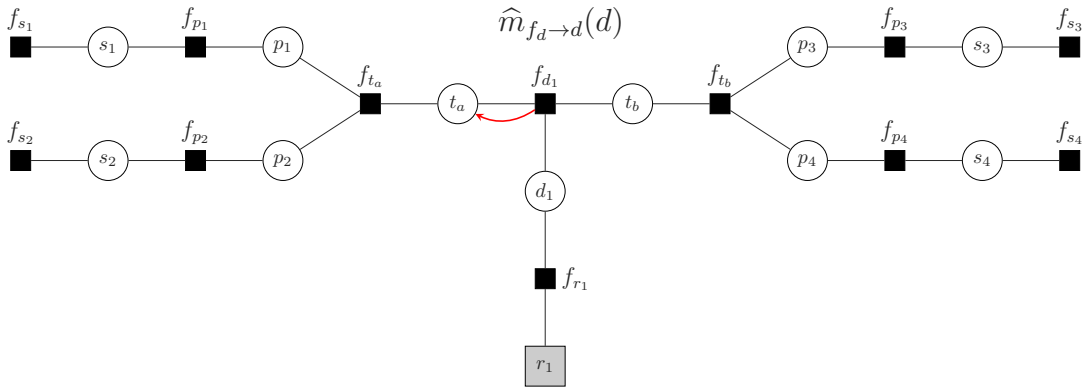
# Mínima Divergencia

Aproximación del posterior de la diferencia

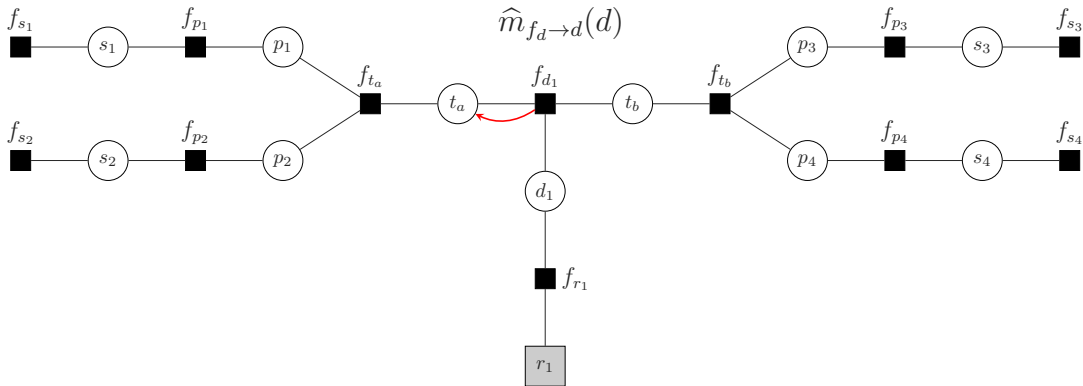
$$p(d, \text{Gana}) = \underbrace{m_{f_d \rightarrow d}(d)}_{P(d)} m_{d \rightarrow f_d}(d)$$

$$\begin{aligned} m_{d \rightarrow f_d}(d) &= \frac{p(d, \text{Gana})}{p(d)} \approx \frac{\hat{p}(d, \text{Gana})}{p(d)} \\ &= \frac{\mathcal{N}(d | \hat{\delta}, \hat{\vartheta}^2)}{\mathcal{N}(d | \delta, \vartheta^2)} \propto \mathcal{N}(d | \delta_{\div}, \vartheta_{\div}^2) \end{aligned}$$

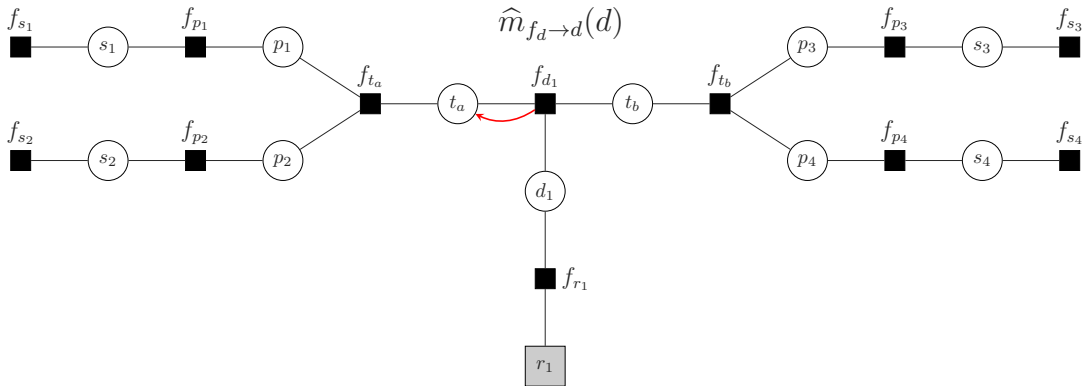
$$\text{con } \delta_{\div} = \frac{\hat{\delta}}{\hat{\vartheta}^2} - \frac{\delta}{\vartheta^2} \text{ y } \vartheta_{\div}^2 = \left( \frac{1}{\hat{\vartheta}^2} - \frac{1}{\vartheta^2} \right)^{-1}$$



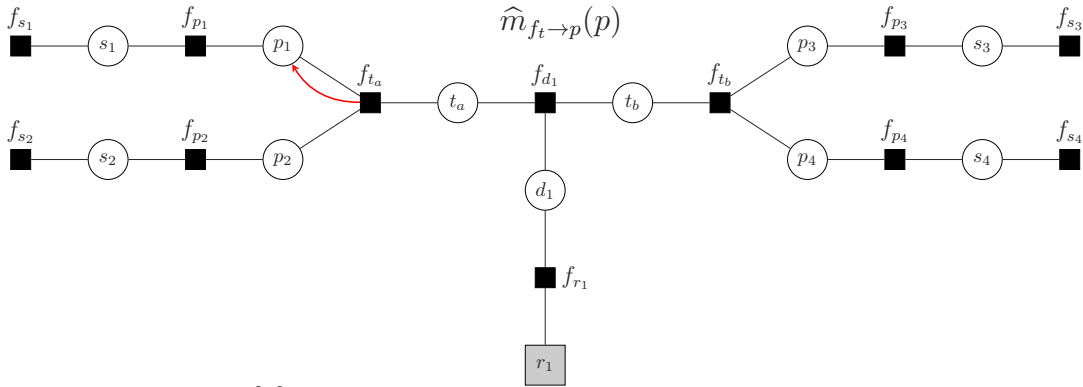
$$\hat{m}_{f_d \rightarrow t_a}(t_a) \approx \iint \mathbb{I}(d = t_a - t_b) \mathcal{N}(d | \delta_{\div}, \vartheta_{\div}^2) \mathcal{N}(t_b | \mu_b, \sigma_b^2) dd dt_b$$



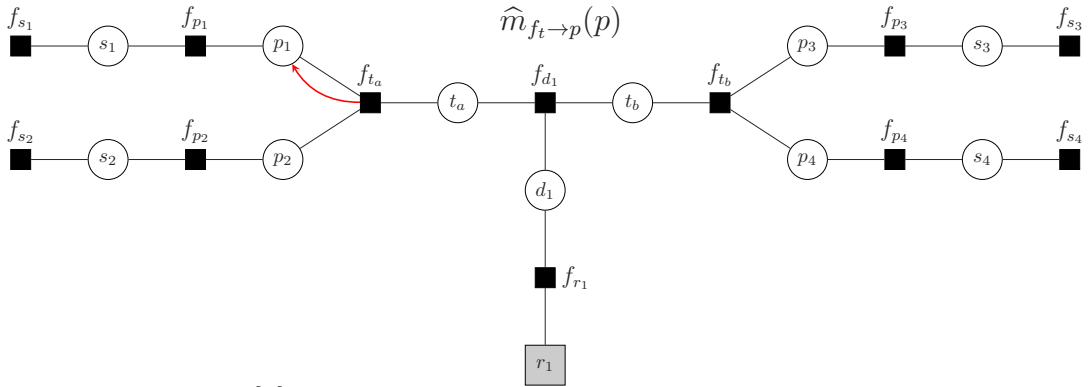
$$\begin{aligned}
 \hat{m}_{f_d \rightarrow t_a}(t_a) &\approx \iint \mathbb{I}(d = t_a - t_b) \mathcal{N}(d | \delta_{\div}, \vartheta_{\div}^2) \mathcal{N}(t_b | \mu_b, \sigma_b^2) dd dt_b \\
 &= \int \mathcal{N}(t_a - t_b | \delta_{\div}, \vartheta_{\div}^2) \mathcal{N}(t_b | \mu_b, \sigma_b^2) dt_b
 \end{aligned}$$



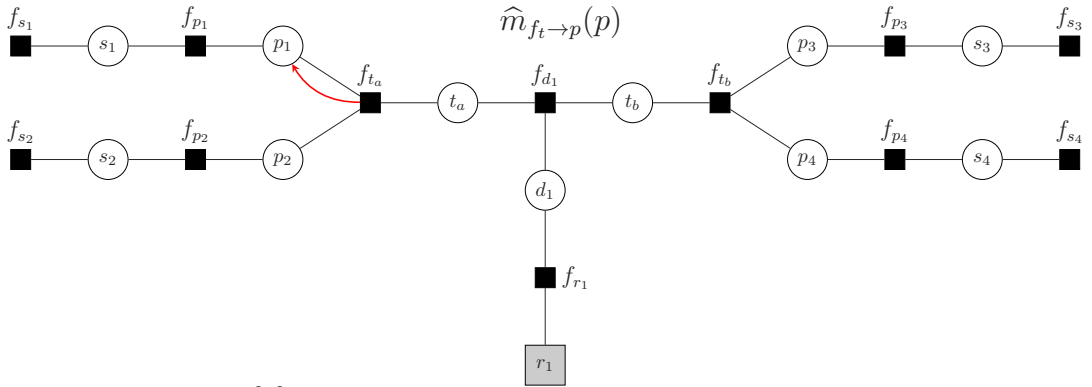
$$\begin{aligned}
 \hat{m}_{f_d \rightarrow t_a}(t_a) &\approx \iint \mathbb{I}(d = t_a - t_b) \mathcal{N}(d | \delta_{\div}, \vartheta_{\div}^2) \mathcal{N}(t_b | \mu_b, \sigma_b^2) dd dt_b \\
 &= \int \mathcal{N}(t_a - t_b | \delta_{\div}, \vartheta_{\div}^2) \mathcal{N}(t_b | \mu_b, \sigma_b^2) dt_b \\
 &= \mathcal{N}(t_a | \mu_b + \delta_{\div}, \vartheta_{\div}^2 + \sigma_b^2)
 \end{aligned}$$



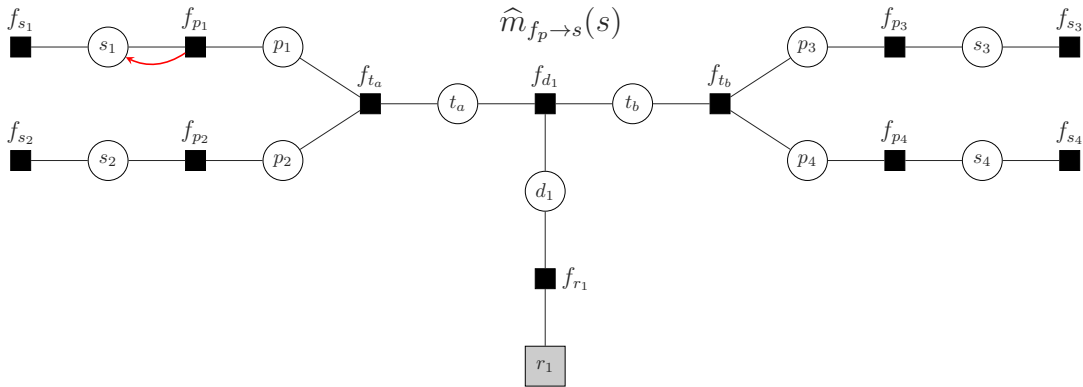
$$\hat{m}_{f_{t_a} \rightarrow p_1}(p_1) = \iint \mathbb{I}(t_a = p_1 + p_2) \mathcal{N}(t_a | \mu_b + \delta_{\div}, \vartheta_{\div}^2 + \sigma_b^2) \mathcal{N}(p_2 | \mu_2, \sigma_2^2 + \beta^2) dt_a dp_2$$



$$\begin{aligned}
 \widehat{m}_{f_{t_a} \rightarrow p_1}(p_1) &= \iint \mathbb{I}(t_a = p_1 + p_2) \mathcal{N}(t_a \mid \mu_b + \delta_{\div}, \vartheta_{\div}^2 + \sigma_b^2) \mathcal{N}(p_2 \mid \mu_2, \sigma_2^2 + \beta^2) dt_a dp_2 \\
 &= \int \mathcal{N}(p_1 + p_2 \mid \mu_b + \delta_{\div}, \vartheta_{\div}^2 + \sigma_b^2) \mathcal{N}(p_2 \mid \mu_2, \sigma_2^2 + \beta^2) dp_2
 \end{aligned}$$

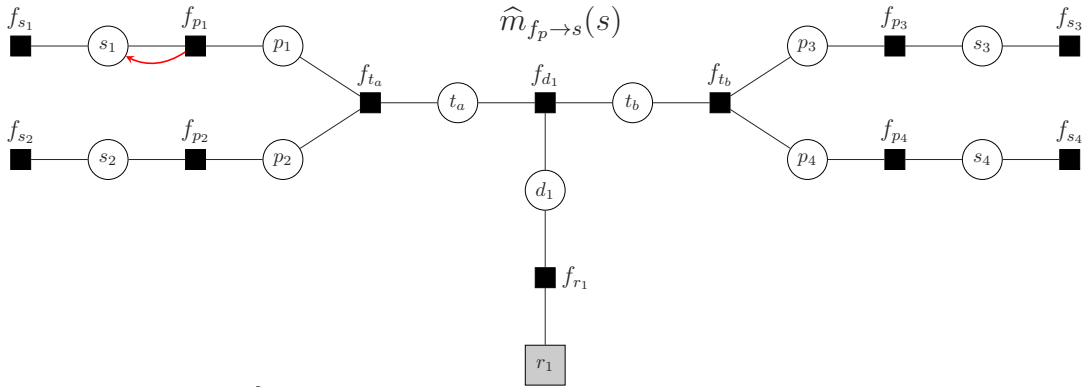


$$\begin{aligned}
 \hat{m}_{f_{t_a} \rightarrow p_1}(p_1) &= \iint \mathbb{I}(t_a = p_1 + p_2) \mathcal{N}(t_a \mid \mu_b + \delta_{\div}, \vartheta_{\div}^2 + \sigma_b^2) \mathcal{N}(p_2 \mid \mu_2, \sigma_2^2 + \beta^2) dt_a dp_2 \\
 &= \int \mathcal{N}(p_1 + p_2 \mid \mu_b + \delta_{\div}, \vartheta_{\div}^2 + \sigma_b^2) \mathcal{N}(p_2 \mid \mu_2, \sigma_2^2 + \beta^2) dp_2 \\
 &= \mathcal{N}(p_1 \mid \underbrace{\mu_b - \mu_2 + \delta_{\div}}_{\mu_1 - \delta}, \underbrace{\vartheta_{\div}^2 + \sigma_b^2 + \sigma_2^2 + \beta^2}_{\vartheta^2 - (\sigma_1^2 + \beta^2)})
 \end{aligned}$$



$$\hat{m}_{f_{p_1} \rightarrow s_1}(s_1) = \int \mathcal{N}(p_1 | s_1, \beta^2) \mathcal{N}(p_1 | \mu_1 - \delta + \delta_{\div}, \vartheta_{\div}^2 + \vartheta^2 - \sigma_1^2 - \beta^2) dp_1$$





$$\begin{aligned}
 \hat{m}_{f_{p_1} \rightarrow s_1}(s_1) &= \int \mathcal{N}(p_1 | s_1, \beta^2) \mathcal{N}(p_1 | \mu_1 - \delta + \delta_{\div}, \vartheta_{\div}^2 + \vartheta^2 - \sigma_1^2 - \beta^2) dp_1 \\
 &= \mathcal{N}(s_1 | \mu_1 - \delta + \delta_{\div}, \vartheta_{\div}^2 + \vartheta^2 - \sigma_1^2)
 \end{aligned}$$

## Posterior aproximado

$$\widehat{p}(s_1, r) = \mathcal{N}(s_1 | \mu_1, \sigma_1^2) \mathcal{N}(s_1 | \mu_1 - \delta + \delta_{\div}, \vartheta_{\div}^2 + \vartheta^2 - \sigma_1^2)$$

# Implementación

# Clase Gaussiana

# Clase Gaussiana

$$\iint \delta(t = x + y) \mathcal{N}(x|\mu_1, \sigma_1^2) \mathcal{N}(y|\mu_2, \sigma_2^2) dx dy = \mathcal{N}(t|\mu_1 + \mu_2, \sigma_1^2 + \sigma_2^2) \quad (\text{N1} + \text{N2})$$

# Clase Gaussiana

$$\iint \delta(t = x + y) \mathcal{N}(x|\mu_1, \sigma_1^2) \mathcal{N}(y|\mu_2, \sigma_2^2) dx dy = \mathcal{N}(t|\mu_1 + \mu_2, \sigma_1^2 + \sigma_2^2) \quad (\text{N1} + \text{N2})$$

$$\iint \delta(t = x - y) \mathcal{N}(x|\mu_1, \sigma_1^2) \mathcal{N}(y|\mu_2, \sigma_2^2) dx dy = \mathcal{N}(t|\mu_1 - \mu_2, \sigma_1^2 + \sigma_2^2) \quad (\text{N1} - \text{N2})$$

# Clase Gaussiana

$$\iint \delta(t = x + y) \mathcal{N}(x|\mu_1, \sigma_1^2) \mathcal{N}(y|\mu_2, \sigma_2^2) dx dy = \mathcal{N}(t|\mu_1 + \mu_2, \sigma_1^2 + \sigma_2^2) \quad (\text{N1} + \text{N2})$$

$$\iint \delta(t = x - y) \mathcal{N}(x|\mu_1, \sigma_1^2) \mathcal{N}(y|\mu_2, \sigma_2^2) dx dy = \mathcal{N}(t|\mu_1 - \mu_2, \sigma_1^2 + \sigma_2^2) \quad (\text{N1} - \text{N2})$$

$$\mathcal{N}(x|\mu_1, \sigma_1^2) \mathcal{N}(x|\mu_2, \sigma_2^2) \propto \mathcal{N}(x|\mu_*, \sigma_*^2) \quad (\text{N1} * \text{N2})$$

# Clase Gaussiana

$$\iint \delta(t = x + y) \mathcal{N}(x|\mu_1, \sigma_1^2) \mathcal{N}(y|\mu_2, \sigma_2^2) dx dy = \mathcal{N}(t|\mu_1 + \mu_2, \sigma_1^2 + \sigma_2^2) \quad (\text{N1} + \text{N2})$$

$$\iint \delta(t = x - y) \mathcal{N}(x|\mu_1, \sigma_1^2) \mathcal{N}(y|\mu_2, \sigma_2^2) dx dy = \mathcal{N}(t|\mu_1 - \mu_2, \sigma_1^2 + \sigma_2^2) \quad (\text{N1} - \text{N2})$$

$$\mathcal{N}(x|\mu_1, \sigma_1^2) \mathcal{N}(x|\mu_2, \sigma_2^2) \propto \mathcal{N}(x|\mu_*, \sigma_*^2) \quad (\text{N1} * \text{N2})$$

$$\sigma_*^2 = \left( \underbrace{\frac{1}{\sigma_1^2}}_{\pi_1} + \underbrace{\frac{1}{\sigma_2^2}}_{\pi_2} \right)^{-1}$$



# Clase Gaussiana

$$\iint \delta(t = x + y) \mathcal{N}(x|\mu_1, \sigma_1^2) \mathcal{N}(y|\mu_2, \sigma_2^2) dx dy = \mathcal{N}(t|\mu_1 + \mu_2, \sigma_1^2 + \sigma_2^2) \quad (\text{N1} + \text{N2})$$

$$\iint \delta(t = x - y) \mathcal{N}(x|\mu_1, \sigma_1^2) \mathcal{N}(y|\mu_2, \sigma_2^2) dx dy = \mathcal{N}(t|\mu_1 - \mu_2, \sigma_1^2 + \sigma_2^2) \quad (\text{N1} - \text{N2})$$

$$\mathcal{N}(x|\mu_1, \sigma_1^2) \mathcal{N}(x|\mu_2, \sigma_2^2) \propto \mathcal{N}(x|\mu_*, \sigma_*^2) \quad (\text{N1} * \text{N2})$$

$$\sigma_*^2 = \left( \underbrace{\frac{1}{\sigma_1^2}}_{\pi_1} + \underbrace{\frac{1}{\sigma_2^2}}_{\pi_2} \right)^{-1} \quad \mu_* = \left( \underbrace{\frac{\mu_1}{\sigma_1^2}}_{\tau_1} + \underbrace{\frac{\mu_2}{\sigma_2^2}}_{\tau_2} \right) \sigma_*^2$$

# Clase Gaussiana

$$\iint \delta(t = x + y) \mathcal{N}(x|\mu_1, \sigma_1^2) \mathcal{N}(y|\mu_2, \sigma_2^2) dx dy = \mathcal{N}(t|\mu_1 + \mu_2, \sigma_1^2 + \sigma_2^2) \quad (\text{N1} + \text{N2})$$

$$\iint \delta(t = x - y) \mathcal{N}(x|\mu_1, \sigma_1^2) \mathcal{N}(y|\mu_2, \sigma_2^2) dx dy = \mathcal{N}(t|\mu_1 - \mu_2, \sigma_1^2 + \sigma_2^2) \quad (\text{N1} - \text{N2})$$

$$\mathcal{N}(x|\mu_1, \sigma_1^2) \mathcal{N}(x|\mu_2, \sigma_2^2) \propto \mathcal{N}(x|\mu_*, \sigma_*^2) \quad (\text{N1} * \text{N2})$$

$$\mathcal{N}(x|\mu_1, \sigma_1^2) / \mathcal{N}(x|\mu_2, \sigma_2^2) \propto \mathcal{N}(x|\mu_{\div}, \sigma_{\div}^2) \quad (\text{N1} / \text{N2})$$

# Clase Gaussiana

$$\iint \delta(t = x + y) \mathcal{N}(x|\mu_1, \sigma_1^2) \mathcal{N}(y|\mu_2, \sigma_2^2) dx dy = \mathcal{N}(t|\mu_1 + \mu_2, \sigma_1^2 + \sigma_2^2) \quad (\text{N1} + \text{N2})$$

$$\iint \delta(t = x - y) \mathcal{N}(x|\mu_1, \sigma_1^2) \mathcal{N}(y|\mu_2, \sigma_2^2) dx dy = \mathcal{N}(t|\mu_1 - \mu_2, \sigma_1^2 + \sigma_2^2) \quad (\text{N1} - \text{N2})$$

$$\mathcal{N}(x|\mu_1, \sigma_1^2) \mathcal{N}(x|\mu_2, \sigma_2^2) \propto \mathcal{N}(x|\mu_*, \sigma_*^2) \quad (\text{N1} * \text{N2})$$

$$\mathcal{N}(x|\mu_1, \sigma_1^2) / \mathcal{N}(x|\mu_2, \sigma_2^2) \propto \mathcal{N}(x|\mu_{\div}, \sigma_{\div}^2) \quad (\text{N1} / \text{N2})$$

$$\sigma_{\div}^2 = \left( \underbrace{\frac{1}{\sigma_1^2}}_{\pi_1} - \underbrace{\frac{1}{\sigma_2^2}}_{\pi_2} \right)^{-1} \quad \mu_{\div} = \left( \underbrace{\frac{\mu_1}{\sigma_1^2}}_{\tau_1} - \underbrace{\frac{\mu_2}{\sigma_2^2}}_{\tau_2} \right) \sigma_{\div}^2$$

# Clase Gaussiana

$$\iint \delta(t = x + y) \mathcal{N}(x|\mu_1, \sigma_1^2) \mathcal{N}(y|\mu_2, \sigma_2^2) dx dy = \mathcal{N}(t|\mu_1 + \mu_2, \sigma_1^2 + \sigma_2^2) \quad (\text{N1} + \text{N2})$$

$$\iint \delta(t = x - y) \mathcal{N}(x|\mu_1, \sigma_1^2) \mathcal{N}(y|\mu_2, \sigma_2^2) dx dy = \mathcal{N}(t|\mu_1 - \mu_2, \sigma_1^2 + \sigma_2^2) \quad (\text{N1} - \text{N2})$$

$$\mathcal{N}(x|\mu_1, \sigma_1^2) \mathcal{N}(x|\mu_2, \sigma_2^2) \propto \mathcal{N}(x|\mu_*, \sigma_*^2) \quad (\text{N1} * \text{N2})$$

$$\mathcal{N}(x|\mu_1, \sigma_1^2) / \mathcal{N}(x|\mu_2, \sigma_2^2) \propto \mathcal{N}(x|\mu_{\div}, \sigma_{\div}^2) \quad (\text{N1} / \text{N2})$$

$$\arg \min_{\mu, \sigma} \text{KL} \left( \underbrace{p(d|\text{Gana})}_{\substack{\text{Gaussiana} \\ \text{truncada}}} || \mathcal{N}(d|\mu, \sigma_2) \right) = \mathcal{N}(d|E(d|\text{Gana}), V(d|\text{Gana})) \quad (\text{N} > 0)$$

# Clase Evento

# Clase Evento

desempeño\_individuos

# Clase Evento

## desempeño\_individuos

$$\begin{aligned} m_{f_{p_i} \rightarrow p_i}(p_i) &= \int \mathcal{N}(p_i | s_i, \beta^2) \mathcal{N}(s_i | \mu_i, \sigma_i^2) ds_i \\ &= \iint \delta(p_i = s_i + o) \underbrace{\mathcal{N}(o | 0, \beta^2)}_{\text{ruido}} \mathcal{N}(s_i | \mu_i, \sigma_i^2) ds_i do \end{aligned}$$

# Clase Evento

## desempeño\_individuos

$$\begin{aligned} m_{f_{p_i} \rightarrow p_i}(p_i) &= \int \mathcal{N}(p_i | s_i, \beta^2) \mathcal{N}(s_i | \mu_i, \sigma_i^2) ds_i \\ &= \iint \delta(p_i = s_i + o) \underbrace{\mathcal{N}(o | 0, \beta^2)}_{\text{ruido}} \mathcal{N}(s_i | \mu_i, \sigma_i^2) ds_i do \end{aligned}$$

```
o = Gaussian(0, 1)
s1 = Gaussian()
p1 = s1 + o
```



# Clase Evento

## desempeño\_equipos

$$m_{f_{t_e} \rightarrow t_e}(t_e) = \iint \delta(t_e = p_i + p_j) \mathcal{N}(p_i | \mu_i, \beta^2 + \sigma_i^2) \mathcal{N}(p_j | \mu_j, \beta^2 + \sigma_j^2) dp_i dp_j$$

# Clase Evento

## desempeño\_equipos

$$m_{f_{t_e} \rightarrow t_e}(t_e) = \iint \delta(t_e = p_i + p_j) \mathcal{N}(p_i | \mu_i, \beta^2 + \sigma_i^2) \mathcal{N}(p_j | \mu_j, \beta^2 + \sigma_j^2) dp_i dp_j$$

ta = p1 + p2  
tb = p3 + p4

# Clase Evento

## diferencia\_equipos

$$m_{f_d \rightarrow d}(d) = \iint \delta(d = t_a - t_b) \mathcal{N}(t_a | \mu_a, \sigma_a^2) \mathcal{N}(t_b | \mu_b, \sigma_b^2) dt_a dt_b$$

# Clase Evento

## diferencia\_equipos

$$m_{f_d \rightarrow d}(d) = \iint \delta(d = t_a - t_b) \mathcal{N}(t_a | \mu_a, \sigma_a^2) \mathcal{N}(t_b | \mu_b, \sigma_b^2) dt_a dt_b$$

d = ta - tb

evidencia = 1.0 - cdf(0, d.mu, d.sigma)

# Clase Evento

marginal\_diferencia

$$\begin{aligned} p(d|\mathbf{Gana}) &\propto \mathcal{N}(d|\psi, \vartheta^2)\mathbb{I}(d > 0) \\ &\approx \mathcal{N}\left(d \mid E(d|d > 0), V(d|d > 0)\right) = \hat{p}(d|\mathbf{Gana}) \end{aligned}$$

# Clase Evento

## marginal\_diferencia

$$\begin{aligned} p(d|\mathbf{Gana}) &\propto \mathcal{N}(d|\psi, \vartheta^2)\mathbb{I}(d > 0) \\ &\approx \mathcal{N}\left(d \mid E(d|d > 0), V(d|d > 0)\right) = \hat{p}(d|\mathbf{Gana}) \end{aligned}$$

$$d_{\text{approx}} = d > 0$$

# Clase Evento

## likelihood\_diferencia

$$p(d, \text{Gana}) = \underbrace{m_{f_d \rightarrow d}(d)}_{\text{Prior}} \underbrace{m_{f_r \rightarrow d}(d)}_{\text{Likelihood}} = p(d)P(\text{Gana}|d)$$

$$\begin{aligned} m_{d \rightarrow f_d}(d) &= m_{f_r \rightarrow d}(d) = \frac{p(d, \text{Gana})}{p(d)} \\ &\propto \frac{p(d|\text{Gana})}{p(d)} \approx \frac{\hat{p}(d|\text{Gana})}{p(d)} \end{aligned}$$

# Clase Evento

## likelihood\_diferencia

$$p(d, \text{Gana}) = \underbrace{m_{f_d \rightarrow d}(d)}_{\text{Prior}} \underbrace{m_{f_r \rightarrow d}(d)}_{\text{Likelihood}} = p(d)P(\text{Gana}|d)$$

$$\begin{aligned} m_{d \rightarrow f_d}(d) &= m_{f_r \rightarrow d}(d) = \frac{p(d, \text{Gana})}{p(d)} \\ &\propto \frac{p(d|\text{Gana})}{p(d)} \approx \frac{\hat{p}(d|\text{Gana})}{p(d)} \end{aligned}$$

$$\text{lh\_d} = \text{d\_approx} / \text{d}$$



# Clase Evento

likelihood\_equipos

$$\hat{m}_{f_d \rightarrow t_a}(t_a) = \iint \underbrace{\delta(d = t_a - t_b)}_{t_a = d + t_b} \mathcal{N}(d_1 | \psi_{\div}, \vartheta_{\div}^2) \mathcal{N}(t_b | \mu_b, \sigma_b^2) dd dt_b$$

# Clase Evento

likelihood\_equipos

$$\hat{m}_{f_d \rightarrow t_a}(t_a) = \iint \underbrace{\delta(d = t_a - t_b)}_{t_a = d + t_b} \mathcal{N}(d_1 | \psi_{\div}, \vartheta_{\div}^2) \mathcal{N}(t_b | \mu_b, \sigma_b^2) dd dt_b$$

$$\text{lh\_ta} = \text{lh\_d} + \text{tb}$$

# Clase Evento

likelihood\_desempeño

$$\hat{m}_{f_{t_a \rightarrow p_1}}(p_1) = \iint \underbrace{\delta(t_a = p_1 + p_2)}_{p_1 = t_a - p_2} \mathcal{N}(t_a | \mu_b + \psi_{\div}, \vartheta_{\div}^2 + \sigma_b^2) \mathcal{N}(p_2 | \mu_2, \sigma_2^2 + \beta^2) dt_a dp_2$$

# Clase Evento

likelihood\_desempeño

$$\hat{m}_{f_{t_a \rightarrow p_1}}(p_1) = \iint \delta(\underbrace{t_a = p_1 + p_2}_{p_1 = t_a - p_2}) \mathcal{N}(t_a | \mu_b + \psi_{\div}, \vartheta_{\div}^2 + \sigma_b^2) \mathcal{N}(p_2 | \mu_2, \sigma_2^2 + \beta^2) dt_a dp_2$$

$$\text{lh\_p1} = \text{lh\_ta} - \text{p2}$$

# Clase Evento

likelihood\_habilidad

$$\hat{m}_{f_{p_1 \rightarrow s_1}}(s_1) = \int \mathcal{N}(p_1 | s_1, \beta^2) \mathcal{N}(p_1 | \mu_{\text{lh}, p_1}, \sigma_{\text{lh}, p_1}^2) dp_1$$

$$\hat{m}_{f_{p_1 \rightarrow s_1}}(s_1) = \iint \underbrace{\delta(p_1 = s_1 + o)}_{s_1 = p_1 - o} \mathcal{N}(o | 0, \beta^2) \mathcal{N}(p_1 | \mu_{\text{lh}, p_1}, \sigma_{\text{lh}, p_1}^2) dp do$$

# Clase Evento

likelihood\_habilidad

$$\hat{m}_{f_{p_1 \rightarrow s_1}}(s_1) = \int \mathcal{N}(p_1 | s_1, \beta^2) \mathcal{N}(p_1 | \mu_{\text{lh.p1}}, \sigma_{\text{lh.p1}}^2) dp_1$$

$$\hat{m}_{f_{p_1 \rightarrow s_1}}(s_1) = \iint \underbrace{\delta(p_1 = s_1 + o)}_{s_1 = p_1 - o} \mathcal{N}(o | 0, \beta^2) \mathcal{N}(p_1 | \mu_{\text{lh.p1}}, \sigma_{\text{lh.p1}}^2) dp do$$

$$\text{lh\_s1} = \text{lh\_p1} - o$$

# Clase Evento

posterior

$$\widehat{p}(s_1, r) = \mathcal{N}(s_1 | \mu_1, \sigma_1^2) \mathcal{N}(s_1 | \mu_{\text{lh.s1}}, \sigma_{\text{lh.p1}}^2)$$

# Clase Evento

## posterior

$$\hat{p}(s_1, r) = \mathcal{N}(s_1 | \mu_1, \sigma_1^2) \mathcal{N}(s_1 | \mu_{\text{lh\_s1}}, \sigma_{\text{lh\_p1}}^2)$$

```
posterior_s1 = s1 * lh_s1
```



$p = \mathbf{b}$

Laboratorios de  
Métodos Bayesianos

# Bibliografía Unidad 4

- Bishop, C. **Pattern Recognition and Machine Learning**. 2006. ([Descargar](#)).  
(lectura 8.1, 8.2 y 8.4)
- Neal. **Pattern Recognition and Machine Learning**. 2020 (Draft). ([Descargar](#)).  
(lectura capítulo 1 y 3)